

Final Project:

Basic Audio Equalizer & Amplifier

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Abstract

Of all of the things electronics now control or are otherwise relied upon in our lives, arguably one of the most commonplace is the speaker; and other audio applications. In order to make a speaker function, an input signal must be taken and run through, most often, an equalizer and an amplifier. The former of the two divides the input signal into 3 different frequency paths, being characterized as low, mid, and high ranges. These are called “low pass”, “band pass”, and “high pass” filters, which as their names describe, allow only low mid or high frequencies to pass through. This is accomplished by first using a combination of resistors and capacitors to filter the frequencies into 3 paths, and then delivering them to their own respective gain control potentiometer. This is what allows individual volume control of each frequency range. Following these gain control knobs, the frequencies are rejoined using a LM324 quad comparator, forming once again, a single signal. This signal is first passed through one final gain control phase, acting as the “master volume” knob, and then pushed through a power amplifier chip, where it is finally sent to the speaker, which in this case presents 8Ω . Being that the speaker is passive instead of active, or internally powered, the signal must first be amplified and powered before being sent to the speaker. At which point it'll have enough power to activate an electromagnet, and rapidly oscillate the speaker's cone to create precise tones, or music. This report will delve into the the design of all 3 previously mentioned frequency filters, their respective cut off requirements, the output V_{RMS} as well as power at minimum and maximum gain for each frequency range, and the ripple measurement of the amplifier.

1. Objectives

The objective of this lab is to create a fully operational and adjustable 3 band equalizer and power amplifier circuit. This will then be used to power and use a speaker, as well as be subjected to tests ensuring proper functionality at each phase of the amplifier's circuitry.

1.1. Filters

Being the first stage of the entire circuit, the frequency pass filters are the first to see the 320 mV RMS, or audio input signal. Using a combination of resistors and capacitors (the values of which will be calculated and further discussed below), the input signal is split into 3 different paths.

The First of these is a low pass filter, which has a cutoff of -3 dB and allows frequencies up to 320 Hz to pass through, for further processing at the gain control knob.

The next filter is the bandpass filter, which similarly has a cutoff value of -3 dB and allows frequencies between 320 Hz and 3200 Hz to pass through. This requires it to either employ RLC, or utilize both a high pass and a low pass filter with a buffer. The latter of the two is the approach used in this report, and an LF356 is employed as the buffer.

Finally, the High Pass Filter, also having a -3 dB cutoff value, allows frequencies of 3200 Hz or greater to pass through, while blocking everything else.

All of these filters then pass their signal to the equalizer section of the circuit for individual adjustment of each frequency range, using an individual operational amplifier for each one. The op amp design for each filter is exactly the same, permitting the use of a LM324 quad comparator chip to handle all three components. All of the technical specifications for each filter can be found in table 1.

1.2. Equalizers

This section of the circuit, as briefly mentioned previously, serves a single purpose. Using an LM324 chip, to play the role of 3 total LF356 IC's, the signal from each of the 3 frequency ranges, is input through a 10k Ω resistor to the negative input terminal of the chip at a node where a 10k Ω potentiometer is also connected. The potentiometer loops around to the output and serves as a feedback loop, allowing volume adjustment for each individual frequency range. It does this by adjusting the resistance ratio of the inverted input signal (which is why the resistor at the input is 10k Ω , to allow a any ratio value from 0 to 1), so that as the potentiometer value increases, so does the gain, and ultimately the volume of that channel. This process is identically repeated for the LPF, HPF, and BPF signals.

1.3. Summing

Upon exiting the equalizer stage of the circuit, the signals are all once again “summed” into one signal, where they will go through one final operational amplifier. This section of the circuit is built in the exact same fashion as each of the 3 individual gain controls, with the exception of the input resistor values. Due to the summing of signals, 3 different signals having a parallel configuration must ultimately sum to $10\text{k}\Omega$ as that is the potentiometer value. To accommodate this, $30\text{k}\Omega$ resistors are used instead. From there the process is identical, as the resistance of the potentiometer increases, so does the gain, and ultimately the volume of the entire system, effectively creating a master volume knob.

1.4. Power Amplifier

The objective of the power amplifier at the very end of the circuit is to ensure the equalizer can output at least 400 mW of power (the necessary amount to power the speaker). It takes the final refined and adjusted voltage from the previous states, and amplifies it using its own power source, before sending it to the speaker where it will become audible noise.

Speaker Impedance	8Ω
Low Pass (Bass) Filter -3dB Cutoff	$<320\text{ Hz}$
Band Pass (Mid) Filter -3dB Cutoff	$320 - 3200\text{ Hz}$
High Pass (Treble) Filter -3dB Cutoff	$>3200\text{ Hz}$
V_{amp} with Max. EQ settings	$< 15\text{ mV}_{RMS} @ 100\text{ Hz} 1000\text{ Hz} 10000\text{ Hz}$
V_{amp} with Min. EQ settings	$100\text{ mV}_{RMS} @ 100\text{ Hz} 1000\text{ Hz} 10000\text{ Hz} \pm 10\%$
Maximum Ripple	15 mV_{RMS}
Output Power Requirement	$> 400\text{ mW} @ 100\text{Hz} - 10\text{kHz}$

Table 1: Audio EQ and Amplifier Specifications

2. Theory

This circuit performs two primary tasks, in which all other functions fall under. The first of these is frequency filtering with gain adjustment, forming an equalizer. The second is powered amplification of the signal, to both produce enough power (400 mW in this case), and deliver a clean (noise-free) signal to the speaker. These will be discussed further below.

2.1 Frequency Filters

In terms of an audio amplifier application such as this one, a frequency filter is nothing more than an organization method which allows isolation of individual frequency ranges for tuning purposes. In this case, 3 different passive filters are implemented, the defining characteristics of which are industry recognized as low pass (bass), band pass (mids), and high pass (treble) filters. The frequency cutoffs can be found in table 1. Once these frequencies are allowed to pass while everything outside of the calculated range for each respective filter is blocked, they can be tuned using a potentiometer feedback loop on an op amp. This will be discussed further in section 2.2.

2.1.1 LPF - Low Pass Filter

The low pass filter, allowing frequencies up to 320 Hz to pass through, consists of a resistor and a capacitor. The capacitor goes to ground, and the output is taken from the node at which the resistor and capacitor are connected. It is the values of these two components that actually determine the cutoff frequency (as seen in equation 1), but the capacitor's path to ground is what creates a short in the circuit for any frequency higher than that of the filter (320 Hz in this case). Essentially, all frequencies are technically allowed in, but once the threshold value is hit, any frequencies higher than that are “discarded” to ground.

In order to calculate the resistor and capacitor values, it is first worth noting that a small resistance value is ideal, as a value that is too high will pose an issue in 2 areas. The first is toggling gain values when only a 10KΩ potentiometer is available, and the second is creating enough power at the output, as a resistance value that is extremely high will limit this maximum value. With this in mind, the following equation is used.

$$f_c = \frac{1}{2 \times \pi \times R \times C}$$

Equation 1: Cutoff Frequency for a Passive Filter

Using our known value of 320 Hz at -3dB as the cutoff frequency, only resistance and capacitance remain. Being that capacitor values offer far less range than resistors, it is wise to choose this value and calculate resistance. In this case a 10uF capacitor is chosen (with the goal of low resistance in mind) and the necessary resistance value is the calculated to be $\approx 49.7359\Omega$ using the following equation.

$$320Hz = \frac{1}{2 \times \pi \times R \times 10\mu F}$$

$$R_{LPF} = 49.73591971\Omega$$

With the resistor value now calculated to be 49.74Ω and the capacitor chosen to be 10uF, the low pass filter is then constructed as follows:

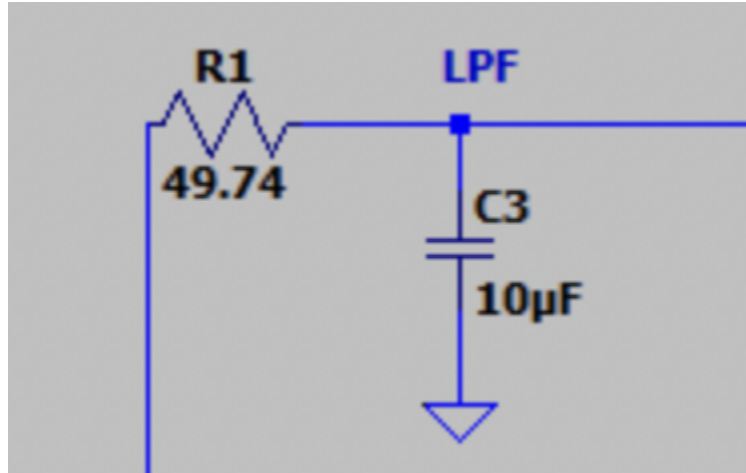


Figure 1: LPF - Low Pass Filter Circuit Schematic

2.1.2 BPF - Band Pass Filter

The band pass filter serves the same purpose as the low pass filter, with the slight difference that it is bounded at both the upper and lower frequencies. Being that it is designed to allow only the middle range of frequencies through, either an RLC circuit, or a combination of two filters (HPF and LPF) with a buffer can be used. The latter approach will be discussed in this report due to limitations imposed by the inclusion of only one inductor in our lab kits.

The frequency cutoff for the band pass filter, also at -3dB, is bounded on the lower end by 320Hz and 3200Hz on the higher end, allowing any frequencies between these two values to pass.

To accomplish this, the low pass filter will be calculated to filter out anything above 3200 Hz and the high pass filter to filter out anything below 320 Hz. In this way, the band pass filter is the exact opposite of the high and low pass filters integrated in this circuit, so as to include the middle range of frequencies. It is worth noting that the high pass filter is constructed in the opposite way to the low pass filter, using a capacitor first with a resistor path to ground. This will be further discussed in section 2.1.3.

These two filters will be “buffered” with an LF356 IC chip, which will minimize interference between the two filters. The capacitor and resistance values for the low pass filter calculated in section 2.1.1 can be used here for the high pass filter, with the configuration of resistor-before-capacitor switched. The values for the low pass filter, still need to be calculated which can be accomplished by solving for the high pass filter, using a frequency cutoff of 3200 Hz at -3dB, and then similarly flipping the configuration of capacitor and resistor. In this case, a 0.1uF capacitor will be used to solve with equation 1, once again aiming for a relatively low resistor value. This yields a resistor value of $\approx 497.359\Omega$

$$3200 \text{ Hz} = \frac{1}{2 \times \pi \times R \times 0.1 \mu\text{F}}$$

$$R_{HPF} = 497.3591971\Omega$$

For consistency, the values taken from the low pass filter of $C=10\mu\text{F}$ and $R=49.74\Omega$ will be converted to use a 0.1uF capacitor as well, providing a new value of $R=4974\Omega$. With these values in mind the band pass filter is constructed like so:

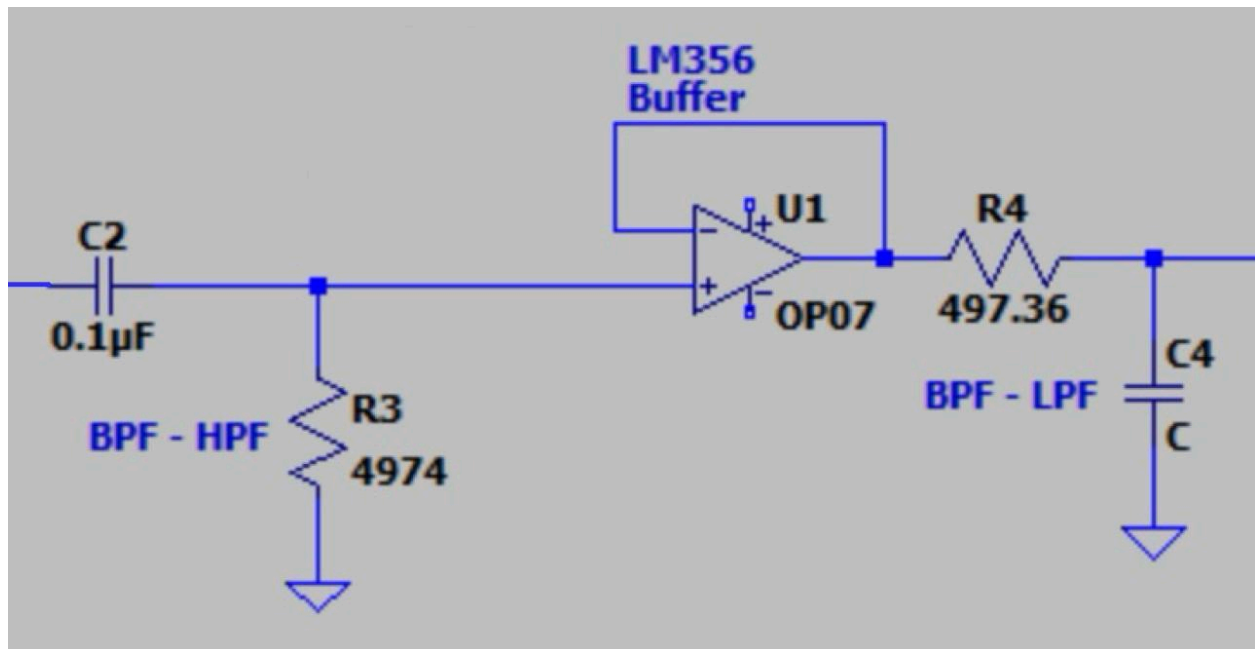


Figure 2: BPF - Band Pass Filter Circuit Schematic

2.1.3 HPF - High Pass Filter

The High pass filter, as described in section 2.1.2, is designed to allow anything from 3200 Hz and above to pass through. Given the necessary resistor and capacitance values have already been calculated, at $C=0.1\mu\text{F}$ and $R=497.36\Omega$, the only remained alteration to the band pass

filter, is to swap the configuration of the capacitor and resistor back so they can once more behave as a HPF and not a LPF.

The high pass filter functions in the exact opposite manner of the low pass filter, this time using a capacitor first, and then a resistor as a path to ground. The output is still taken from the node at which the two are connected, but due to the capacitor taking its effect first, the opposite filter behavior is observed. As all frequencies enter, only those *above* the cutoff frequency are allowed to pass as a result of this flipped configuration.

A table of all filters component values can be found below in section 2.1.4. The High Pass filter is designed as such:

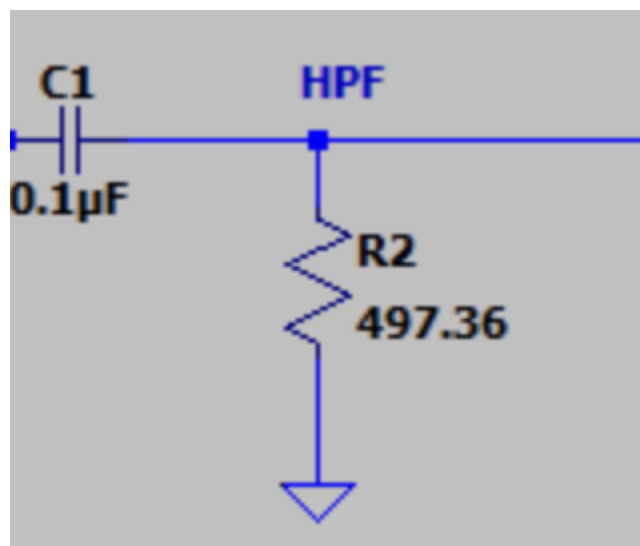


Figure 3: HPF - High Pass Filter Circuit Schematic

2.1.4 Filter Component Values

	$R_{\text{calculated}} (\Omega)$	$R_{\text{applied}} (\Omega)$	C (uF)
LPF - Low Pass Filter	49.735	47	10
HPF - High Pass Filter	497.359	495	0.1
BPF - Low Pass Filter	497.359	495	0.1
BPF - High Pass Filter	4973.591	4950	0.1

Table 2: Resistor and Capacitor Values for Each Frequency Pass Filter

2.2 Powered Amplification | Operational Amplifier

Operational amplifiers makeup the two final and only undiscussed components of this circuit. They appear in very different ways, but each operate very similarly. Being active components, these chips require independent input voltage (in some cases positive and negative), to then amplify and alter the signal with which they're working. Their primary purpose in this particular circuit is to provide gain control, and to amplify the final output before sending it to a speaker.

2.2.1 Gain Control

The first and most frequent occurrence of an operational amplifier, servicing 4 separate frequency channels, is the LM324, serving the purpose of gain control. When the LM324 (in this case) chip is implemented alongside a feedback loop using a potentiometer, a gain/volume control knob is effectively the result. In all applicable locations throughout this circuit, the input equivalent resistance to the chip is 10KΩ to ensure an adjustable ratio/gain from 0 to 1 using the 10kΩ potentiometer. It is worth noting that a single LM324 chip is used instead of 3 different LF356's, while 3 different potentiometers are still used (for each frequency filter). A single LF356 and only one potentiometer are used to control the summed signal, for ease of assembly purposes. Using the following equation:

$$Gain = \frac{R_{potentiometer}}{R_{input}}$$

Equation 2: Gain Calculation for Feedback Loop Application

It can be surmised that as the value of the potentiometer increases, so will the gain, eventually reaching a value of 100%, or "1". This implementation is designed as seen in figure 4, with little to no difference among each of the 4 applications found in figure 6.

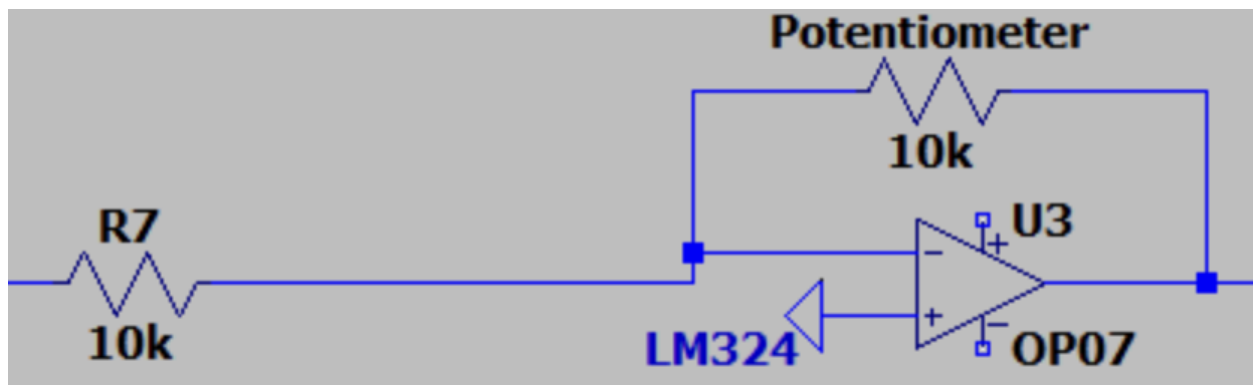


Figure 4: Gain Adjustment Circuit Schematic Using LM324 Op Amp

2.2.2 Output Amplification

The second operational amplifier to be used in this circuit is the LM 386. This chip's implementation and operation is relatively simple. With an internal gain of up to 20, and the ability to drive loads ranging from 4Ω - 32Ω , it has no issue accomplishing the necessary output power of 400mW for this 8Ω speaker application. It functions by taking its supply voltage, and using it to amplify the input signals amplitude. It then outputs the amplified signal directly to the speaker, where it is ultimately converted to sound. The schematic for this chip's integration can be found in figure 5, but it is worth noting, that as with previous schematics in this writeup, the power supply is not depicted. Opposed to the positive and negative voltage required by the LM324 chips ($\pm 6V$ in this case), the LM386 requires only a positive input voltage, the value of which directly affects the amplification coefficient (+6V for testing).

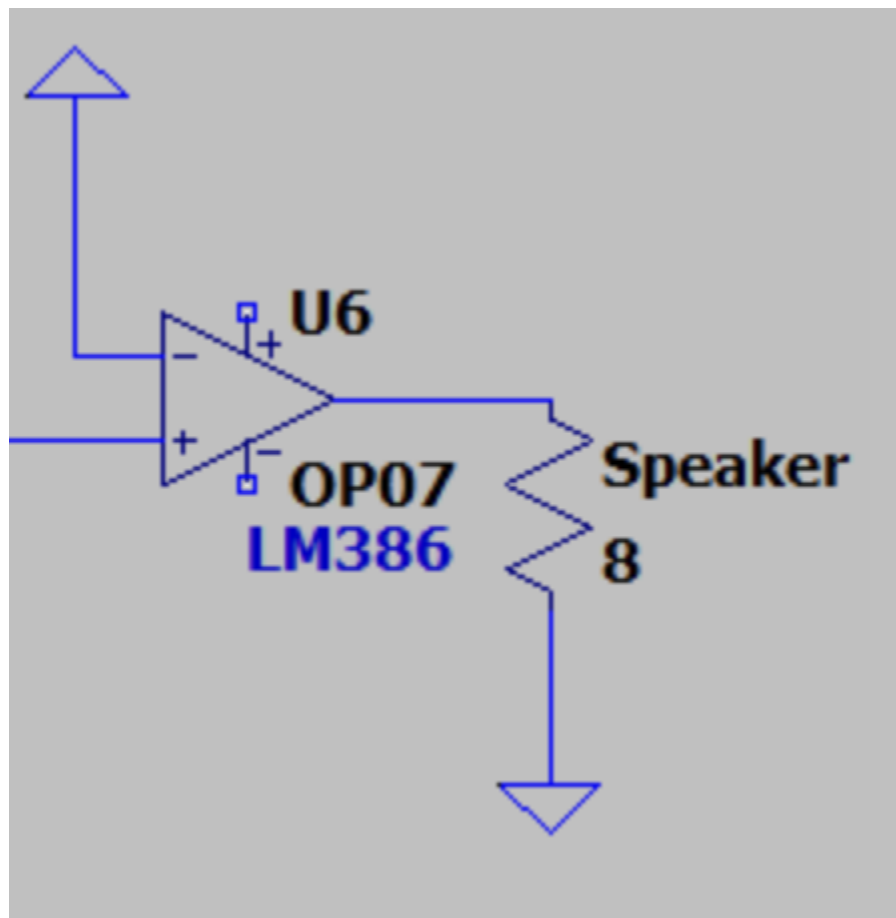


Figure 5: Power Amplifier Output Circuit Schematic

3. Procedure

The general procedure for creating each phase of this circuit, the phases being frequency filtering, equalizing/gain adjustment, and summing / power amplifier output adjustment, were relatively similar. For all applicable components, such as resistors and capacitors, values needed to first be calculated. This is already thoroughly discussed in section 2. They were then implemented, and tested as a standalone device (a filter by itself for example), to confirm functionality without lots of noise or interference introduction. From there, the next component is implemented, tested, and then tied in with the previous component until the entire circuit is built and tested at each step for functionality. All components share a common ground, and the power application is discussed in section 2.2.2. The equalizer & amplifier circuit in its entirety can be found in figure 6.

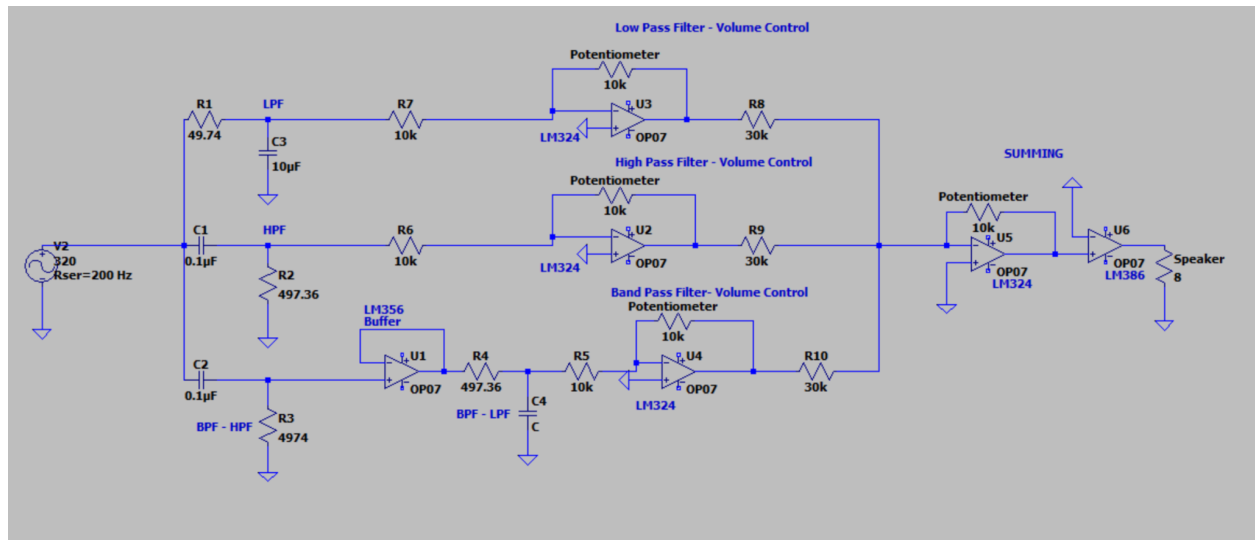


Figure 6: Audio Equalizer & Amplifier Circuit Schematic

Testing for all components, throughout the entire procedure of building the circuit, were conducted using a 320 mVrms Hi-Z sin wave, at varying frequencies as needed (based on which filter, if any, was being tested).

3.1 Frequency Filters

During the testing of each filter, when testing V_{amp} , a 320 mVrms Hi-Z sin wave, at 100, 1000, or 10000 Hz was applied, based on the use of the LPF, BPF, or HPF. When testing the functionality of each filter however, an FRA spanning an appropriate frequency range was run, with a pass criteria of $\pm 10\%$ of the target frequency at -3dB for each test.

3.1.2 LPF - Low Pass Filter

As was outlined in section 2.1.1, the procedure for constructing the low pass filter involves choosing a capacitor value, in this case a 10uF capacitor, and calculating for resistance using the frequency cutoff equation.

$$f_c = \frac{1}{2 \times \pi \times R \times C}$$

That value in this case, was found to be 49.74Ω, with 47Ω chosen to be accurate enough for application. With step one complete, the filter is constructed on a breadboard as follows:

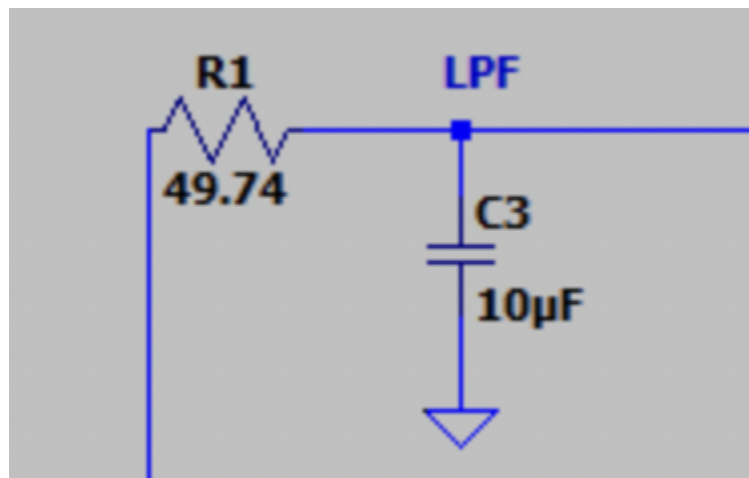


Figure 7: LPF - Low Pass Filter Circuit Schematic

It is then tested by running an FRA (frequency response analysis), over the course of 10Hz - 1kHz, being that the cutoff value is expected to fall at 320 Hz, at -3dB. The results of this test will be discussed further in section 4. Finally, the filter is finalized, and set up permanently on the breadboard, awaiting integration to future components.

3.1.2 BPF - Band Pass Filter

Calculating the necessary values to construct the band pass filter conveniently also provided the values necessary to construct the high pass filter, as discussed in section 2.1.2. Just as with the low pass filter, the frequency cutoff equation is used to calculate the low pass filter resistor requirement, after choosing a 0.1 uF capacitor.

$$f_c = \frac{1}{2 \times \pi \times R \times C}$$

The value for the high pass filter integrated into the band pass filter, comes directly from the low pass filters calculations, with a reversal of component integration in its circuit schematic. It is worth noting at this point, that for the sake of consistency among the band pass filter, the

capacitor value taken from the LPF (and used for the band pass filters HPF), was changed to 0.1 uF and the resistor value is adapted accordingly. The product of these calculations can be found in table 3.

	$R_{calculated} (\Omega)$	$R_{applied} (\Omega)$	C (uF)
BPF - Low Pass Filter	497.359	495	0.1
BPF - High Pass Filter	4973.591	4950	0.1

Table 3: Resistor and Capacitor Values Band Pass Filter

With step one complete, just as with the LPF, the band pass filter is constructed on a breadboard using an LF356 IC as a buffer.

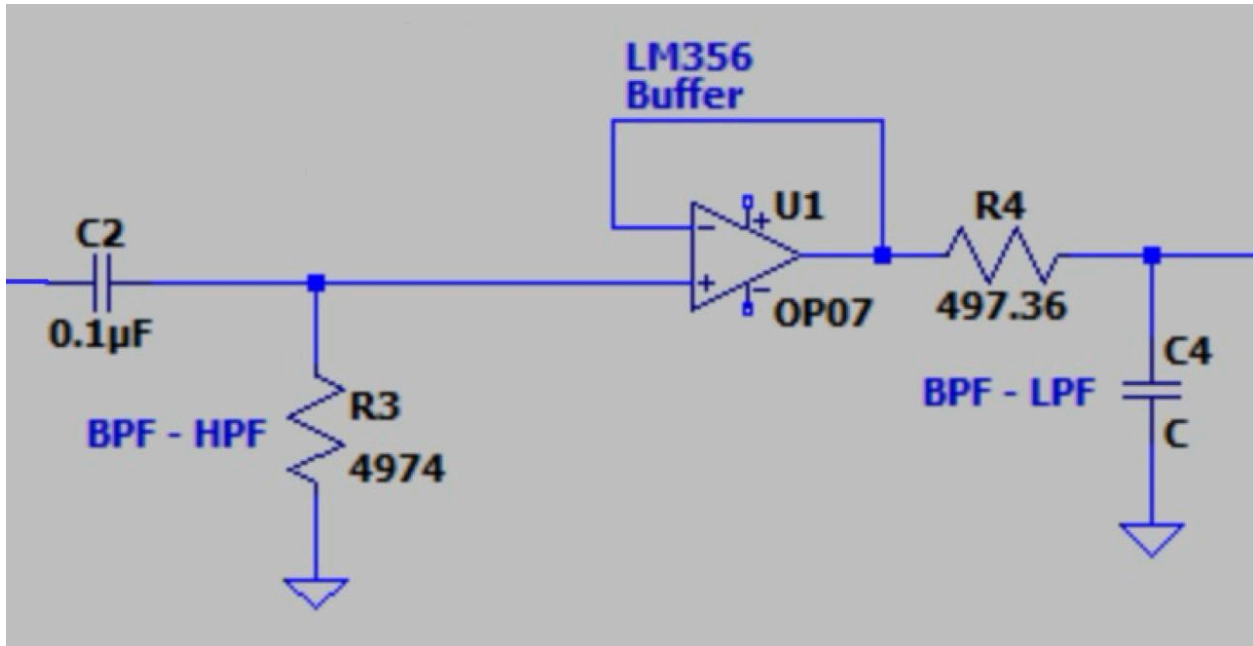


Figure 8: BPF - Band Pass Filter Circuit Schematic

The Buffer assembly is relatively simple, as it uses a feedback loop to first take in the high pass filter signal, and loop it back around from the output to the negative terminal of the chip. From there it will then be smoothly and stably integrated into the low pass filter out of the output pin again, and eventually moved along to the equalizer. This is done by feeding the signal from the HPF into pin 3, adding a short circuit between pins 6 and 2, and then running the signal through the LPF following the second exit of pin 6. Additionally, the chip needs +V and -V to function, applied at pins 7 and 4, respectively. 6V was used in this case.

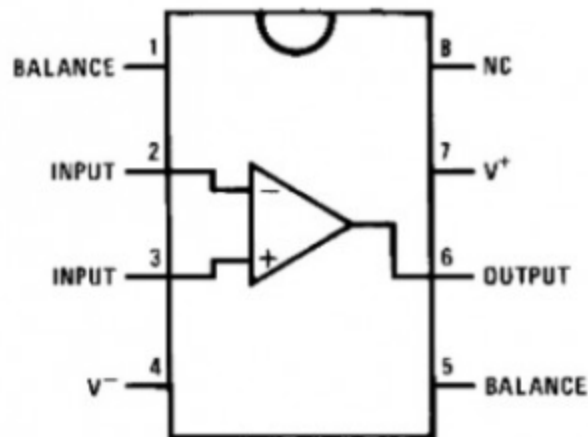


Figure 9: Pinout Diagram for LF356 IC

Following its construction, it also undergoes testing in the form of an FRA (frequency response analysis), with applied frequencies ranging from 10Hz - 10kHz. These results will be discussed in depth in section 4. Once functionality is confirmed, with frequency readings of 320 Hz and 3200 Hz $\pm$ 10% at -3dB, the filter is finalized and the next one is tested.

3.1.2 HPF - High Pass Filter

As was outlined in section 2.1.1, the procedure for constructing the high pass filter involves choosing a capacitor value, in this case 0.1uF is chosen for the sake of achieving a low resistor value. At this point however, the calculation has already been performed for the resistor value of the LPF within the band pass filter, and all that is left to do is reverse the location of the resistor and capacitor.

This resistance value was found to be 497.35 Ω , with 495 Ω chosen to be sufficiently adequate for application. With step one complete, the filter is constructed on a breadboard as follows:

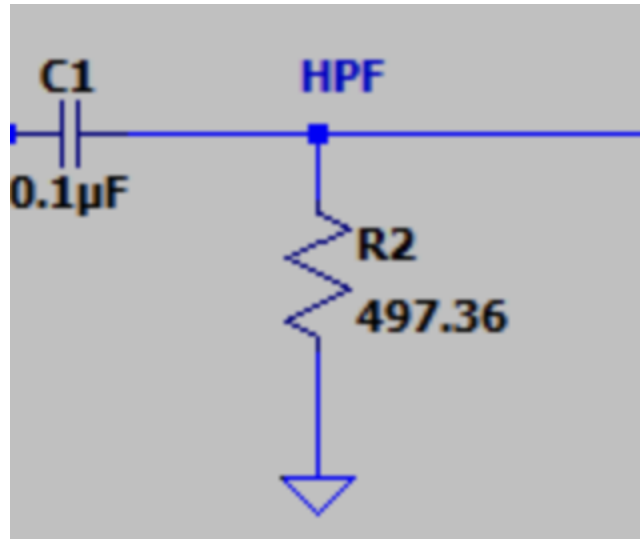


Figure 10: HPF - High Pass Filter Circuit Schematic

It is then tested by running an FRA (frequency response analysis), over the course of 2kHz - 10kHz, being that the cutoff value is expected to fall at 3200 Hz, at -3dB. The results of this test will be discussed further in section 4. Finally, the filter is finalized, and set up permanently on the breadboard, awaiting integration to the equalizer section of the amplifier.

3.2 Operational Amplifiers

Following the frequency filters, op amps are responsible for most of if not the rest of the functionality of this circuit. The first application of them occurs while the signal is still isolated into its low mid and high channels, providing gain control of each individual frequency range. They are then implemented with the purpose of recombining the signals, and then again for the task of amplifying the signal enough to send to a speaker.

3.2.1 Gain Control / EQ Operational Amplifier

To accomplish gain control, a circuit must first be designed using a feedback loop, and a potentiometer integrated into the short between output and input. With that in mind, the input resistance as previously discussed must match that of the potentiometer, to create a perfect range from 0-1 gain (or 0-100%). This is covered in more detail in section 2.2.1, where a schematic of said feedback loop can be found (figure 4). With that said, the equalizer section of the full schematic found in figure 6 must be further discussed.

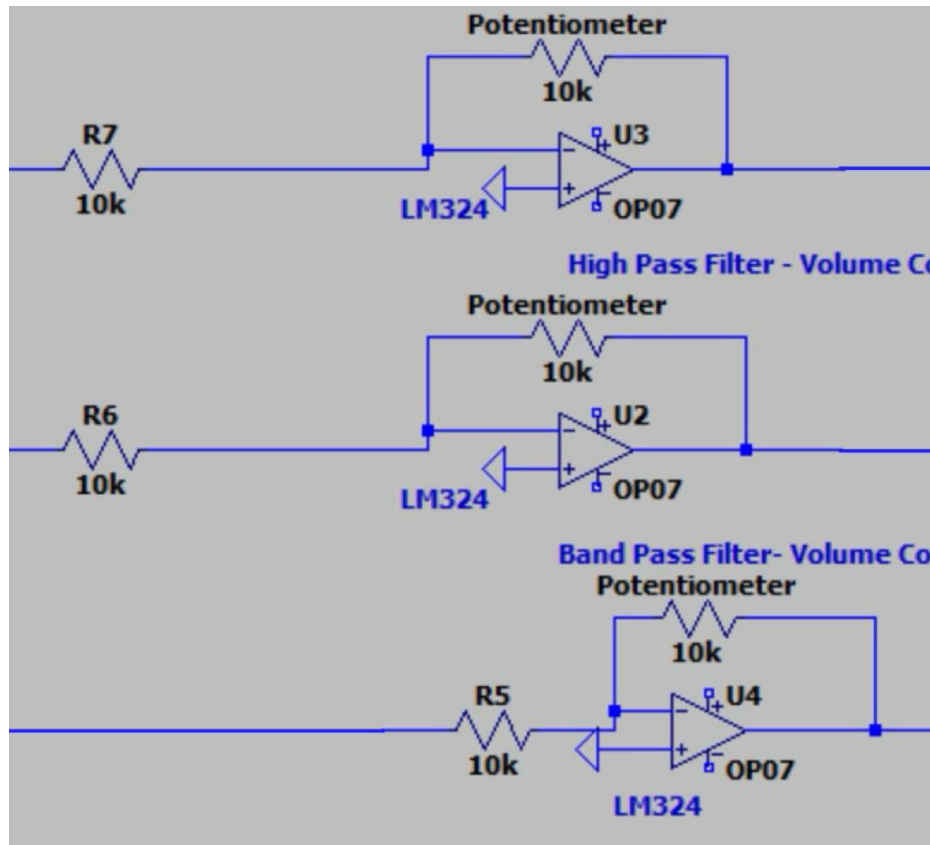


Figure 11: Gain Control For Each Frequency Range

It is worth noting that this schematic can be practically applied using 3 different LF 365 chips or a single LM324 chip, with one of its comparators unused. The LM324 is what's used here in an inverting setup. It is inverted by taking the input signal through its input resistor (10kΩ for all channels), and feeding it first into the negative input of the chip, while the positive input is grounded. In similar fashion to the application presented 2.2.1, the output is then processed and found at the output of the chip, where it is affected by the feedback loop, creating the ratio of loop to input resistance, i.e: gain.

Just as with each frequency filter, although there is very little to calculate here (only resistance matching), once the circuit is constructed on the breadboard, and integrated into their respective filters, they are tested at the output for functionality. The test this time, does not involve an FRA, but rather live analysis of oscilloscope readings as the potentiometer is toggled, with a 320 mVrms sine wave input. The frequency of this input is either 100Hz, 1kHz, or 10kHz, depending on which filter is being tested, to ensure the frequency actually makes it to the equalizer.

The construction of the LM324 circuit is nearly identical to that found in figure 9 for the LF356, although there are 3 of them.

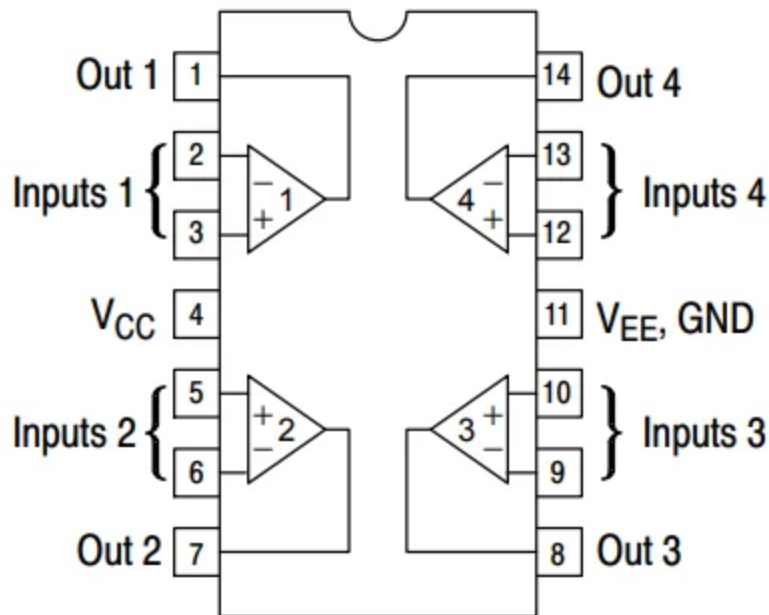


Figure 12: LM324 Operational Amplifier Pinout

With that in mind, to actually construct the circuit, the output signal from each respective frequency filter, is first run through a 10k Ω resistor, and then into pins 2, 6, and 13. The order of this does not matter, as the application is identical. Pins 3, 5, and 12 are all directly grounded, and from pins 2 to 1, 13, to 14, and 6 to 7, there is a connection made with potentiometer, or: a feedback loop. These 3 potentiometers are what offer full variable gain control over each frequency channel. Finally, channels 4 and 11 receive +6V and -6V.

This is how a single LM324 is used to create gain and volume control over each frequency range, forming an equalizer.

3.2.2 Summing & Powered Amplification

The last task to be completed both within the circuit and by an op amp, is to sum the 3 signals back into one, and then run it through a power amplifier before delivering it to the speaker. The first is a generally simple task, and operates exactly the same way as the equalizer. Using an LM324 chip, (the same one if desired, only 1 of the 4 op amp's is necessary. An LF356 would have worked just as well), all 3 outputs from the equalizer are joined at a single node, and then run into the negative input of the op amp. In the same fashion as the equalizer, the output of the chip is looped back to the negative input through a potentiometer. Being that all signals have now been rejoined, this creates a "master" gain control knob, effecting what is now the only, cumulative signal in the circuit.

The difference at this stage lies within the input resistance. Due to component accessibility restrictions, a 10kΩ potentiometer must still be used. With that in mind, the output of each of the 3 frequency channels before summing, is now 10kΩ as well. Being that they are in parallel, the equivalent resistance must be calculated.

$$\frac{1}{R_{EQ}} = \frac{1}{R_{LPF}} + \frac{1}{R_{BPF}} + \frac{1}{R_{HPF}}$$

$$\frac{1}{R_{EQ}} = \frac{1}{10k} + \frac{1}{10k} + \frac{1}{10k}$$

$$R_{EQ} = 30k\Omega$$

With this value and its necessity in mind, 30kΩ resistors are added before the summing node on all 3 channels, to ensure that the input resistance into the final gain control, is equal to the potentiometer value of 10kΩ, creating one final instance of a 0-1 gain range.

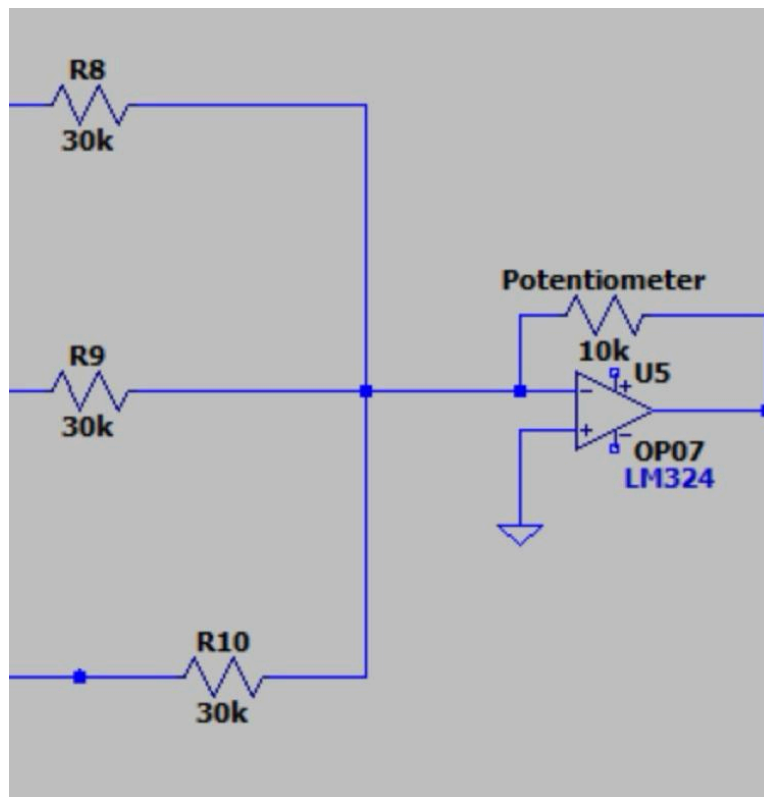


Figure 13: Signal Summing and Resistance Adjustment

With these adjustments aside, the setup from there on is identical to the previous description. Pins 4 and 11 receive +6V and -6V respectively, the output is taken from the output &

potentiometer node, and the positive pin (of whichever of the 4 op amps is being used) is grounded to create an inverted application.

When all of this has been sorted out, as per the general procedure, this circuit is implemented on the breadboard, and tested *with* integration into the rest of the circuit. Standalone testing is not necessary at this juncture unless troubleshooting, because the circuit itself has already been tested and is now reliant upon the signal from the previously tested components. Demonstrations of low, medium and high volume, as a result of toggling this gain knob, can be found with a further discussion in section 4.

3.2.3 Powered amplification

The final point of activity for operational amplifiers in this circuit is the LM386 power amplifier, used to amplify the finally processed signal enough to output to a speaker. Its circuit construction is arguably the simplest of them all, albeit critical. It is determined that a gain of 20, the chip's internal default is sufficient for this application, so its dedicated gain pins (1 and 6) do not require any signal.

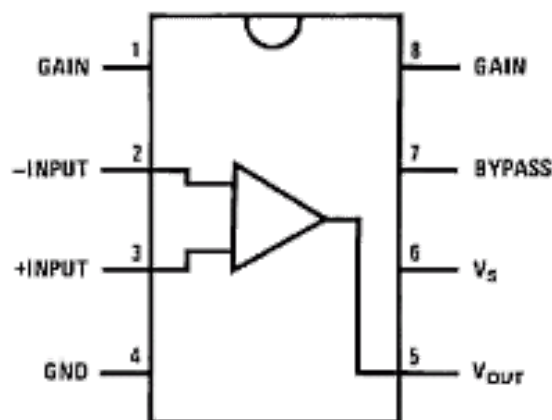


Figure 14: LM 386 Power Amplifier Pinout

The chip is implemented in a non-inverted fashion, opposing the rest of the circuit, and is given +6V to its V_s terminal at pin 6. In the same vein, pins 4 and 2 are both grounded, the output from the summing amplifier is sent to pin 3, and eventually output from pin 5. Pin 7 is left alone.

The operation of this chip is very simple, requiring no loops or external control to perform its task. It simply uses its independent supply voltage, to amplify the input signal, and output it with a gain of 20 (unless otherwise specified), to a load that must fall between 4Ω and 32Ω . The speaker being used is 8Ω , which does not pose an issue. Furthermore, it does not require a negative voltage. The output from this chip is the final destination of the circuit, as it is then sent to the speaker.

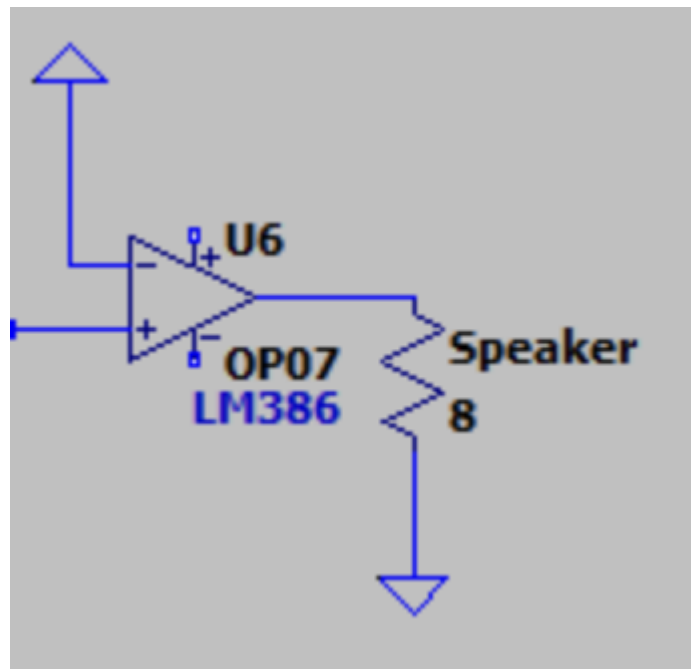


Figure 15: LM 386 Power Amplifier Circuit Schematic

As per procedure, this is next to be tested, which is performed using the ripple, output power, and output V_{RMS} values as testing criteria. These will all be further discussed in section 4.

4. Results

Many of the acquired results have been alluded to or received promise of further discussion throughout this report. The following results verify benchmarks found in table 4, which indicate proper speaker and amplifier function including the following:

Speaker Impedance	8 Ω
Low Pass (Bass) Filter -3dB Cutoff	<320 Hz
Band Pass (Mid) Filter -3dB Cutoff	320 - 3200 Hz
High Pass (Treble) Filter -3dB Cutoff	>3200 Hz
V_{amp} with Max. EQ settings	< 15 mV_{RMS} @ 100 Hz 1000 Hz 10000 Hz
V_{amp} with Min. EQ settings	100 mV_{RMS} @ 100 Hz 1000 Hz 10000 Hz $\pm$ 10%
Maximum Ripple	15 mV_{RMS}
Output Power Requirement	> 400 mW @ 100Hz - 10kHz

Table 4: Audio EQ and Amplifier Specifications

Tests have been performed to check and verify each of these benchmarks are achievable. The first of these criteria is not one to be tested, but rather a foundational value for which to build the circuit, stating that the speaker itself is indeed, 8 Ω .

4.1 Filters

The first 3 of these benchmarks to be tested are the low pass, high pass, and band pass filters. As stated in section 3.1, these filters must display an output of 320 Hz, 3200 Hz, or both at -3dB with a $\pm$ 10% tolerance. This is tested by running an FRA on each filter.

4.1.1 Low Pass filter

The results of the low pass filter, as seen in figure 16, indicate a signal of 335Hz at -2.97dB. This was the closest measurement automatically taken during the FRA to -3dB, so its value can be considered true, as long as the percent error does not approach 10%. The cutoff frequency for this filter is 320 Hz. With that said, the percent error from the target specification is as follows:

$$\%_{error} = \left| \frac{335-320}{335} \right| \times 100$$

$$\%_{error} = 4.47\%$$

Being that 4% is well under the 10% tolerance for all filters, the low pass filter has performed quite successfully.

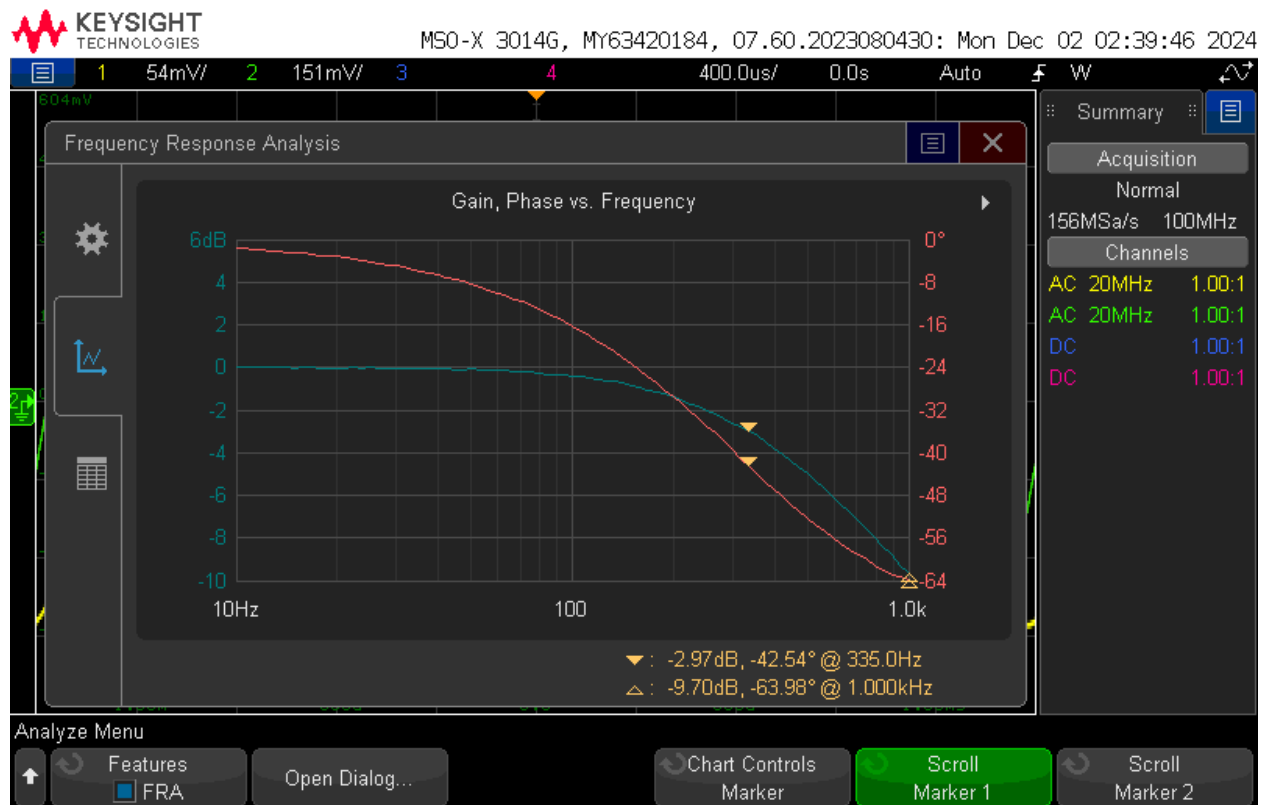


Figure 16: Low Pass Filter FRA

4.1.2 Band Pass filter

The FRA run on the band pass filter, spanning from 10 Hz to 10kHz, which is notably excessive as the upper bound of the filter is set at 3200 Hz, can be found in figure 17. This FRA poses a similar interpretation issue as the LPF FRA. To amend this, measurements have been documented manually for the low pass filter, as the FRA jumps from -2.89dB to -3.24dB. With that said, at -2.89 decibels, the frequency reading is 316Hz, and as seen in figure 17 at -3.24dB, the frequency reading is 290.1Hz. For the sake of confirmation, a proportional middle ground is found, at -3dB and 307.86Hz. With a target cutoff frequency of 320 Hz, the percent error is as follows:

$$\%_{error} = \left| \frac{320 - 307.86}{320} \right| \times 100$$

$$\%_{error} = 3.79\%$$

The tolerance for both sections of the band pass filter is also 10%, so the low pass filter is functioning exactly as intended.

The high pass filter on the other hand, is far easier to comprehend, providing a readout at -3.01dB, which can be considered ≈ 3 . At 2.985kHz, the percent error is:

$$\%_{error} = \left| \frac{3.2 - 2.985}{3.2} \right| \times 100$$

$$\%_{error} = 6.718\%$$

Although this value is higher than that of the LPF, it is still within the $\pm 10\%$ error range for frequency cutoffs. The band pass filter is functioning comfortably within its requirement found in table 4.

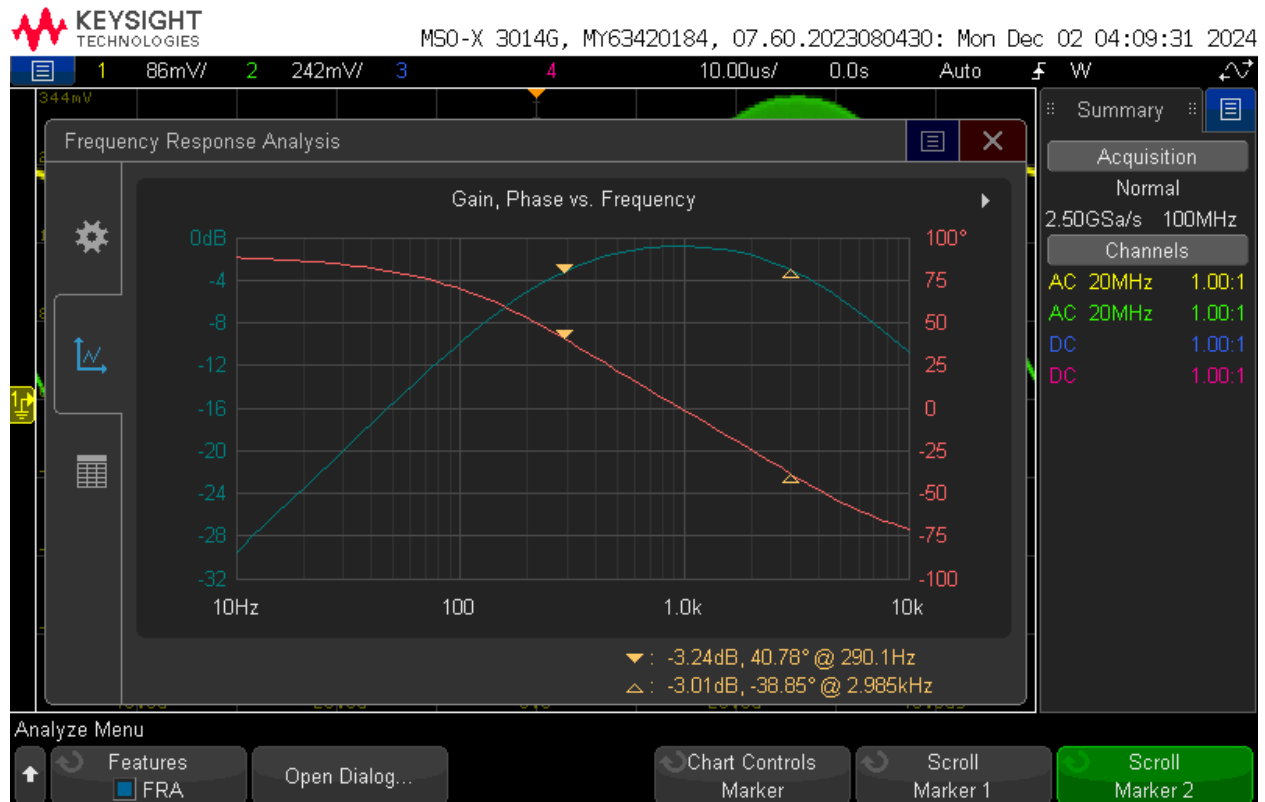


Figure 17: Band Pass Filter FRA

4.1.3 High Pass filter

Just as with the low and band pass filters, the high pass filter's functionality is tested by running an FRA, this time from 2kHz to 10kHz. Conveniently, a recording is taken at -3dB exactly, eliminating the need for regression. With a target cutoff frequency of 3200 Hz, the real value is found to be 3241Hz.

$$\%_{error} = \left| \frac{3.2 - 3.241}{3.2} \right| \times 100$$

$$\%_{error} = 1.281\%$$

With a 1.281% error, the high pass filter is the most accurate filter of the 3, and also well within its 10% tolerance range. These results are seen in figure 18.

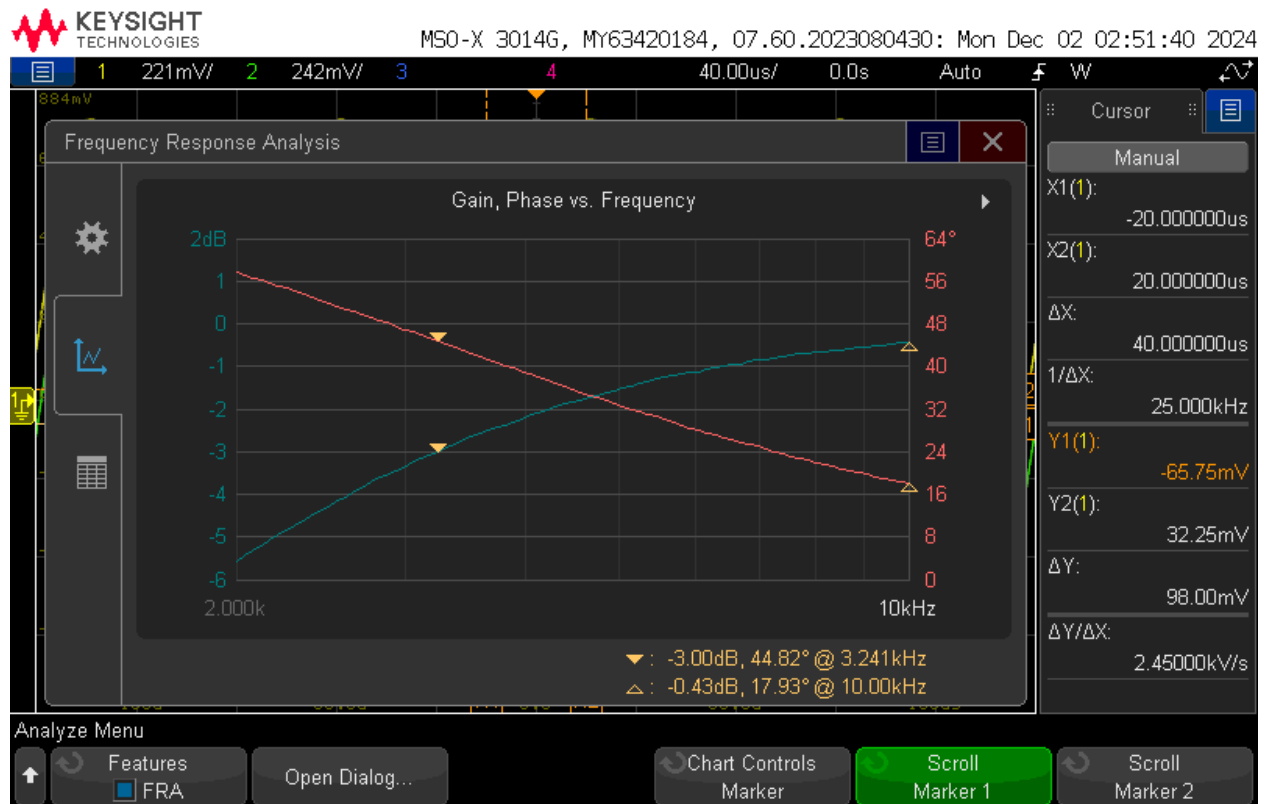


Figure 18: High Pass Filter FRA

4.2 Operational Amplifiers

The next two primary stages of the circuit, as touched on in section 3.2, are reliant on some type of operational amplifier. In this case the LM324 is used for its integral 4-in-1 design, and the LM386 is used for its amplification ability. Both are tested individually and with different methods, to be discussed below.

4.2.1 Summing Amplifier

It has been already discussed why and how the LM324 chip is implemented at the summing stage within this circuit. With that said, in order to test its functionality, it is critical to test the final singular output with all equalizers set to maximum (to ensure they work at the top end of their gain range, and with all set to minimum (to ensure they work at the lower end of their gain range). Furthermore, in order to confirm that each filter itself is working properly, a test at both of these criterias must be performed 3 times, at 100 Hz (to target the LPF), 1000 Hz (to target the BPF) and at 10,000 Hz (to target the HPF). At minimum gain, the target output reading is 15 mVrms or less because it should be 0, and at maximum gain the target output is 100 mVrms because this is the input to the powered amplifier and a 400 mW output from the powered amplifier is desired. All tests are performed at 320 mVrms input.

4.2.1.1 Minimum Gain

The first test is performed at minimum gain, targeting the LPF with an input frequency of 100Hz, found in figure 19.

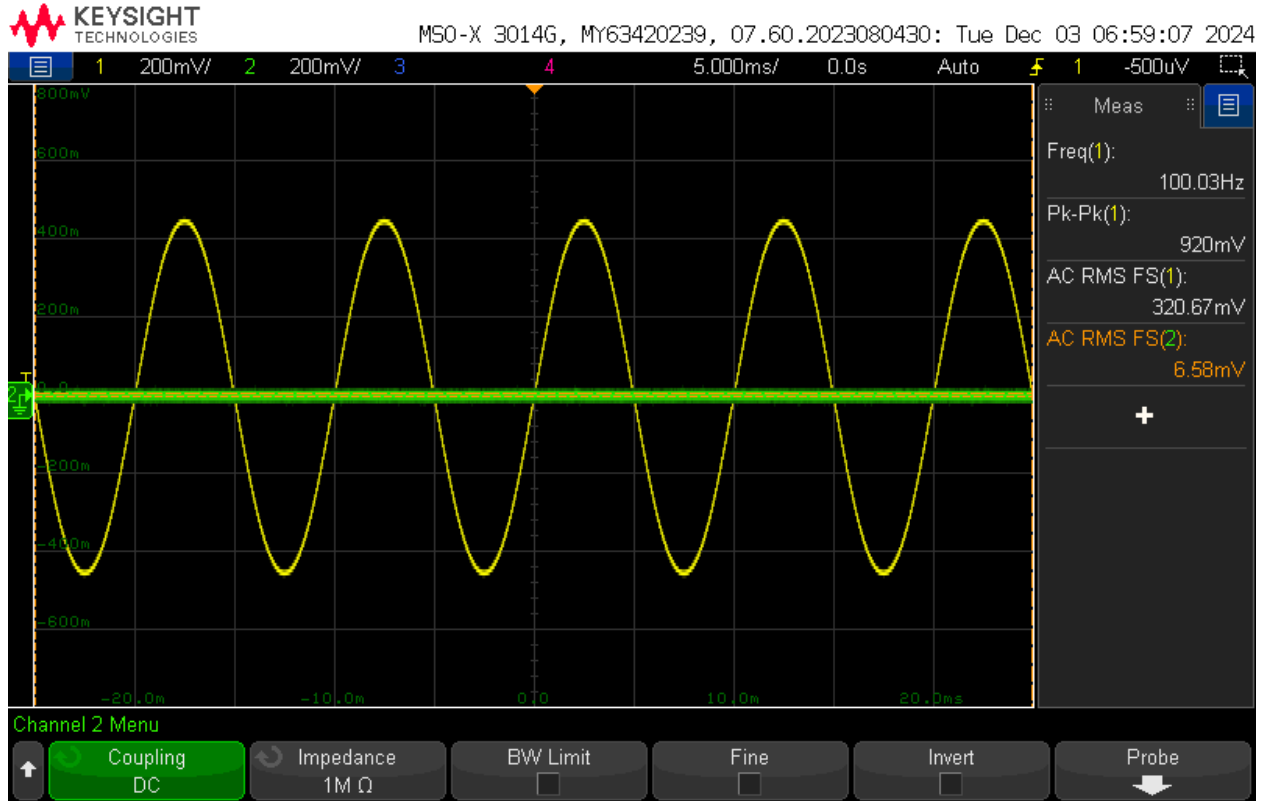


Figure 19: AC RMS Output with 100Hz Input at Minimum Gain

The input signal is taken as well for verification of clean signal input, and the output is measured at 6.58 mVrms, which is well under the 15 mVrms threshold, confirming proper functionality.

The second test, also performed at minimum gain, targets the band pass filter this time using an input signal of 1000 Hz, which falls within the BPF's 320-3200 Hz range. The result can be found in figure 20.

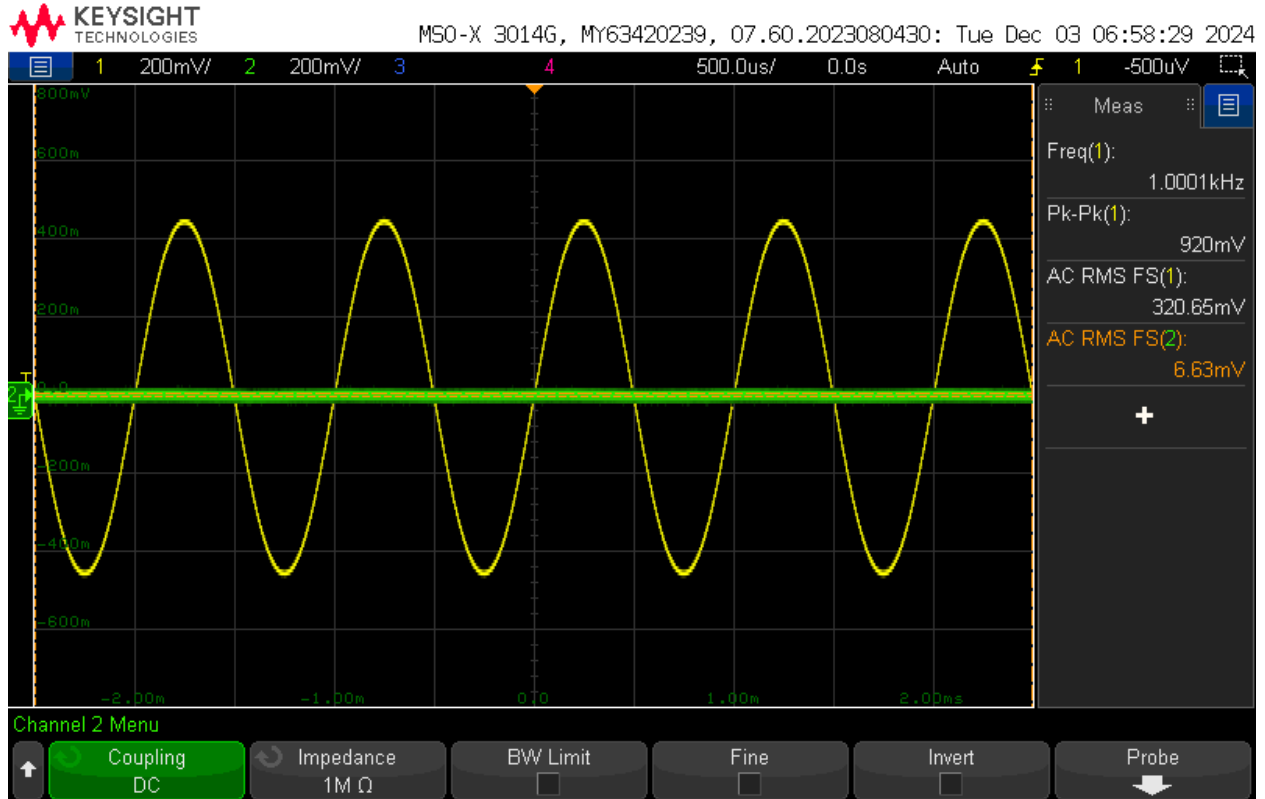


Figure 20: AC RMS Output with 1000 Hz Input at Minimum Gain

Once more, the output falls extremely close to zero at 6.63 mVrms this time, which is also well under the 15 mVrms specified threshold. The band pass filter is functioning properly.

Finally, the High pass filter is tested at minimum gain, with an input frequency of 10kHz, as seen in figure 21.

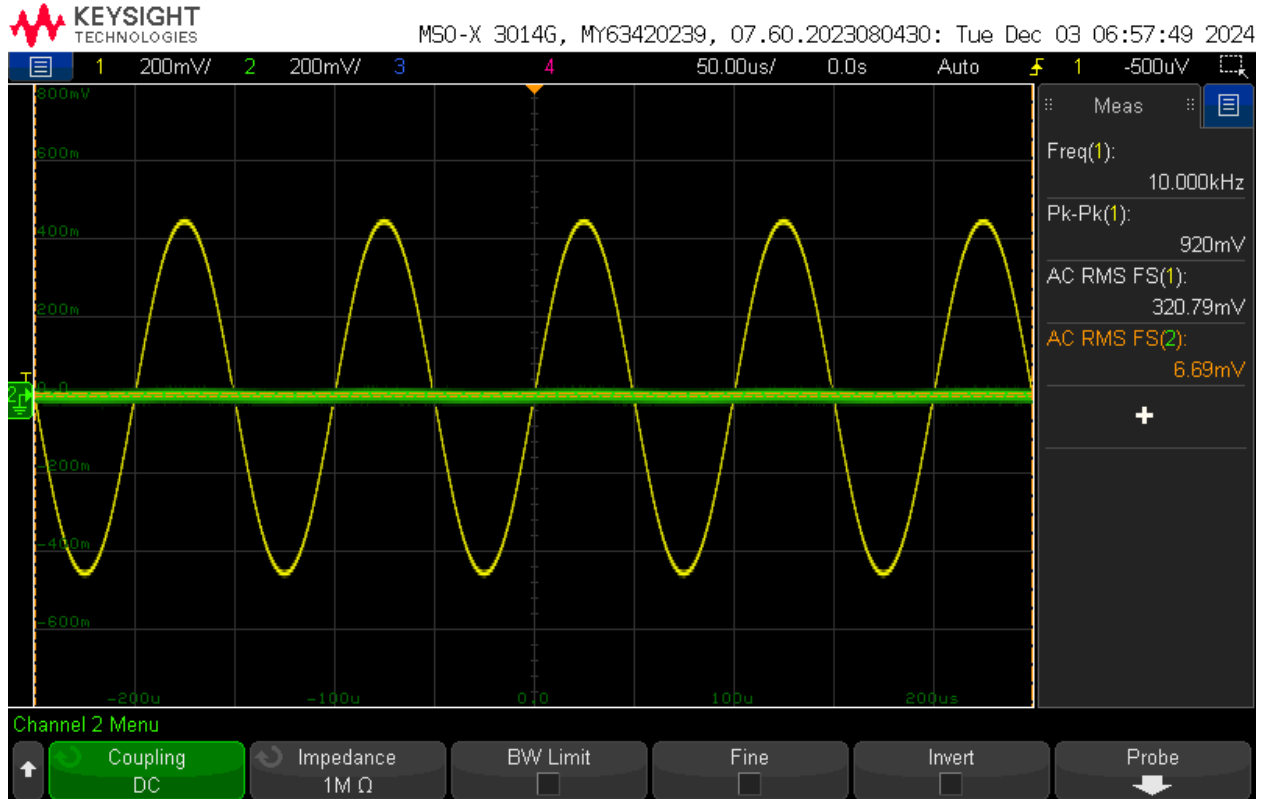


Figure 21: AC RMS Output with 10000 Hz Input at Minimum Gain

With the output being 6.69 mV this time, and relatively consistent in regard to noise, the high pass filter also provides an output well under the threshold of 15 mVrms. This indicates that at minimum gain, all filters are working properly after being re-joined.

4.2.1.2 Maximum Gain

An identical set of tests is performed with all potentiometers set to 10k Ω , or all gains set to maximum. The first of which is with a 100 Hz input signal targeting the LPF, with a target output of 100 mVrms. The result of this test is found in figure 22.

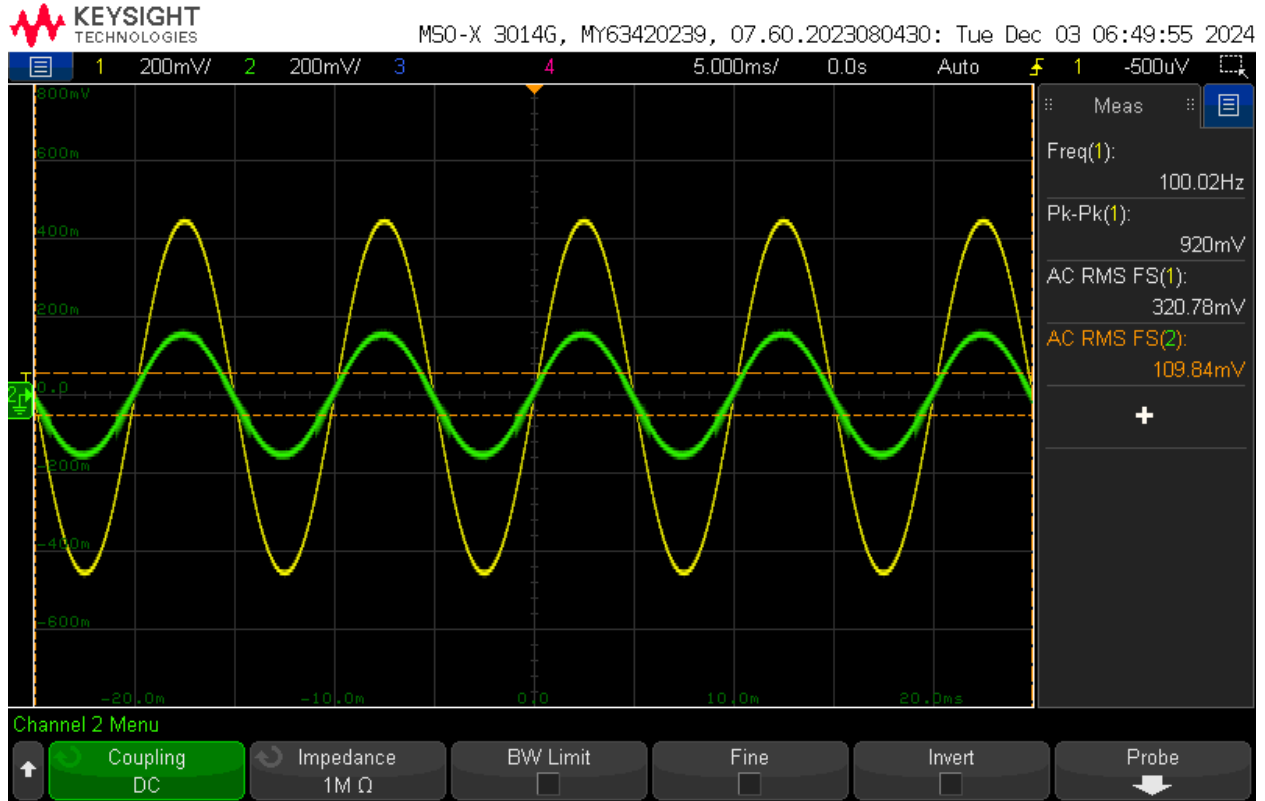


Figure 22: AC RMS Output with 100Hz Input at Maximum Gain

Given the specification found in table 4 for maximum gain output is a value and not a range below a certain number (i.e. 0-15 mVrms), a tolerance must be applied to ensure all components are working as they are intended. A $\pm 10\%$ tolerance will be applied here as well, for the sake of consistency. With that said, the output at 100Hz yields 109.84 mVrms, which is 9.84% off, and within the 10% error range.

The next test is performed at 1000 Hz, and can be found in figure 23.

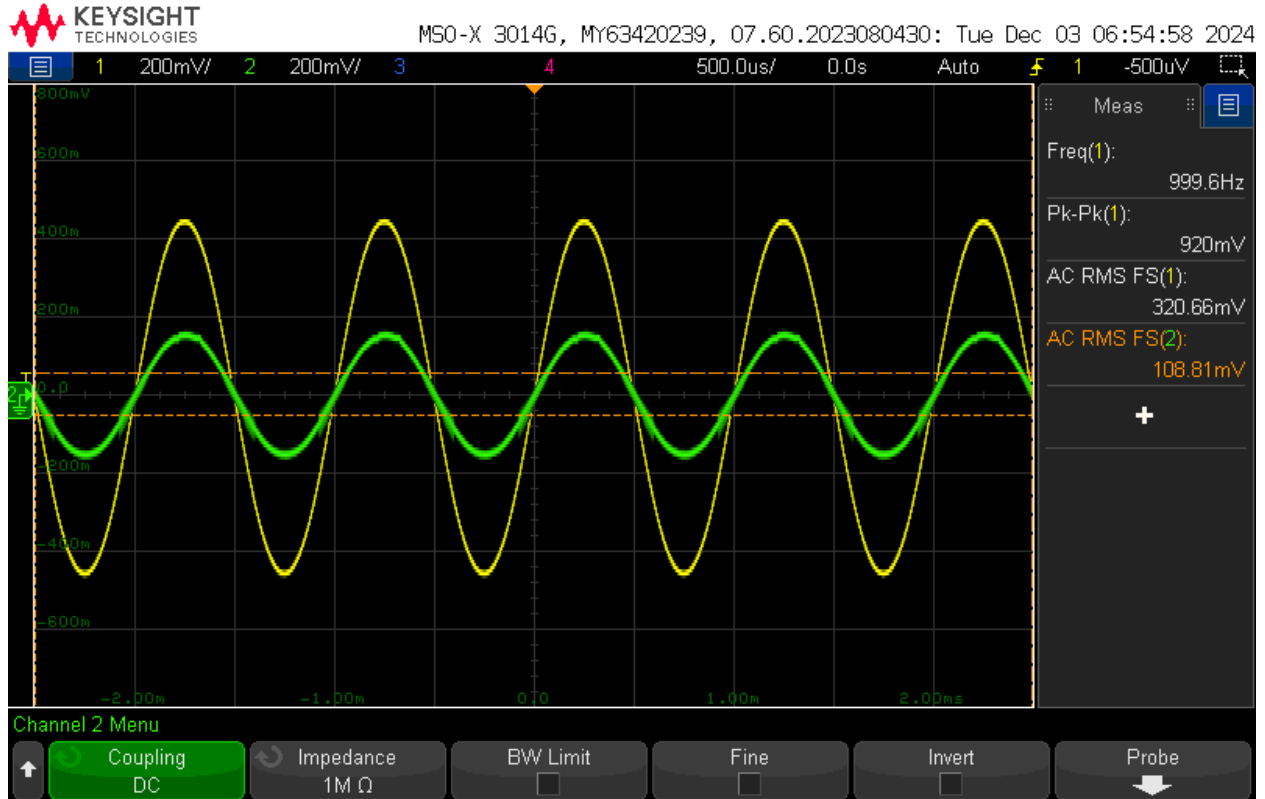


Figure 23: AC RMS Output with 1000 Hz Input at Maximum Gain

Similarly, the output when testing with 1kHz, slightly overshoots the target value of 100mVrms, at 108.81 mVrms. With an 8.81% error though, it is also within the acceptable range and is confirmed to be functioning properly.

Finally, a test at maximum gain is performed at 10kHz, targeting the HPF. The result of this test is found in figure 24.

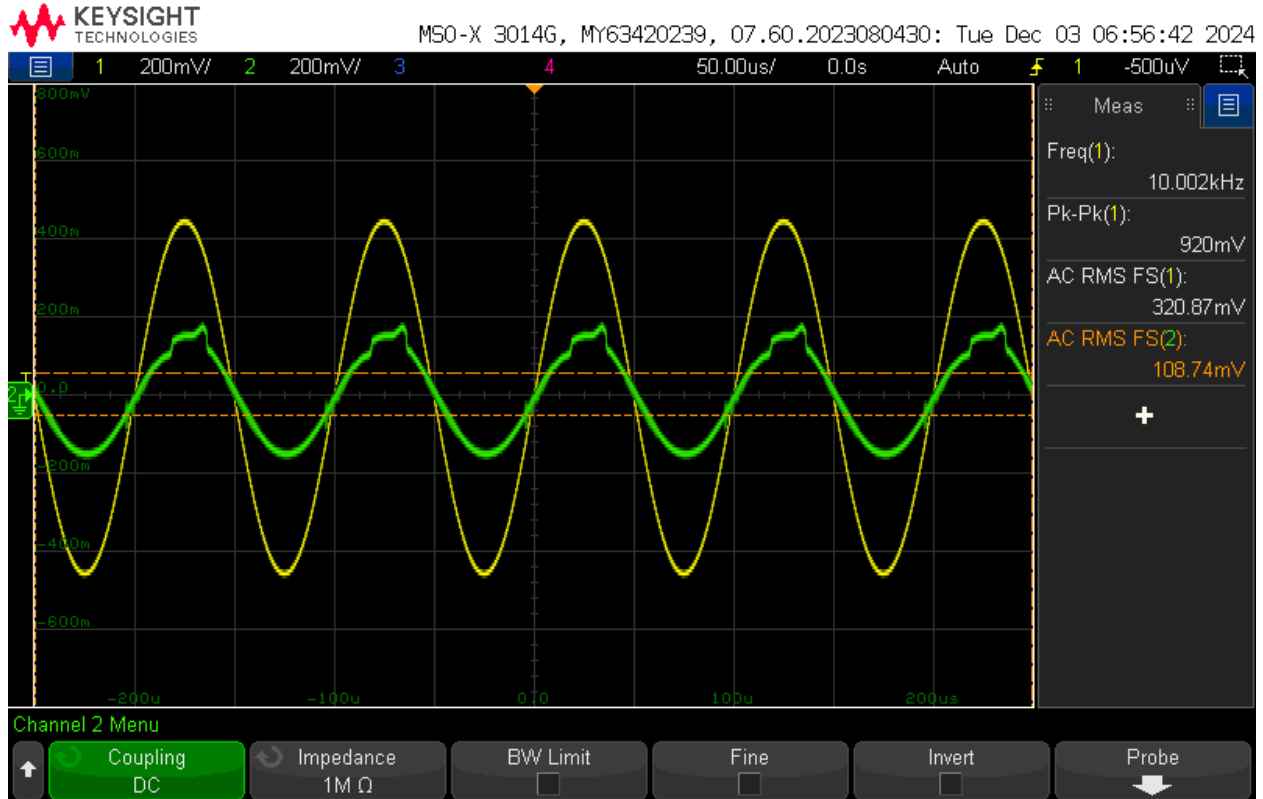


Figure 23: AC RMS Output with 10000 Hz Input at Maximum Gain

This output has some obvious deformations at its peak, which were likely a result of faulty connection, and did introduce noise to the system. A decoupling capacitor was added later and this issue was amended. The output value though, of 108.74 mVrms, yields an 8.74% error, indicating that it is also functioning properly at maximum gain.

It is worth noting at this point, that the applied resistance values for all filters, shown in table 2, is slightly less in each instance than the exact value calculated. This is likely responsible for the over-shot output values during each maximum gain test.

All things considered, the summing amplifier performs exactly as expected under all input and EQ configurations.

4.2.1.3 Ripple Measurement

The final test to ensure the summing amplifier is working as specified, is the ripple measurement. In theory, the ripple is a simple measurement, the steps of recording which are to run an FRA, and then note the minimum and maximum frequencies of the output. From there the oscilloscope is used to measure AC RMS values at the output, while a waveform generator inputs each of the previously recorded min and max frequencies. The ripple is then calculated by subtracting these voltages.

$$Ripple = V_{RMS,Max} - V_{RMS,Min}$$

Equation 3: Ripple Calculation

The ripples target value as specified in table 4 is 15 mVrms or less, which indicates a very low amount of noise and unwanted variation.

First, the FRA is run, as seen in figure 24. From this test, the minimum and maximum frequencies are found to be 100Hz, and 768.6 Hz. These are determined by finding the extreme dB locations throughout the test. The next step is to power the circuit with 100 Hz and 768.6Hz sine waves, both at 320 mVrms, and measure the output, as seen in figures 24 and 25.

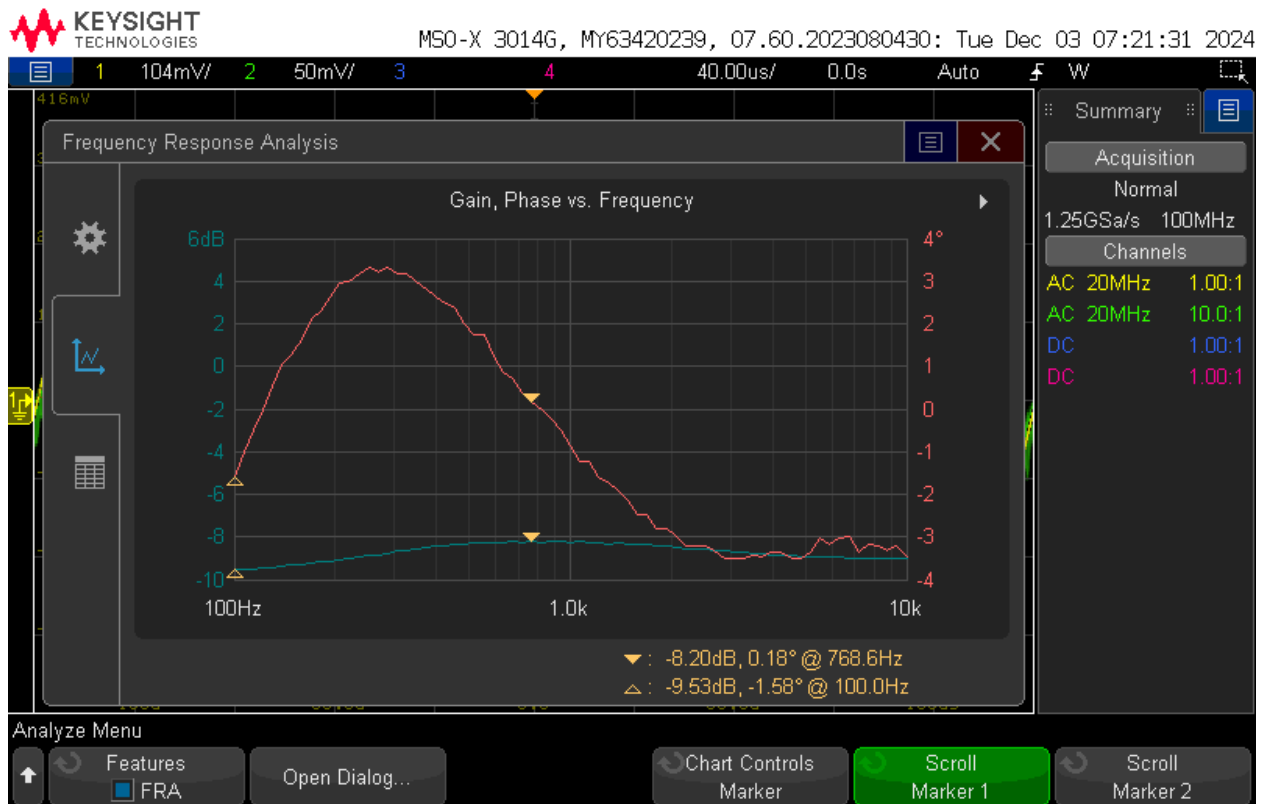


Figure 23: Ripple FRA At Summing Amplifier Output

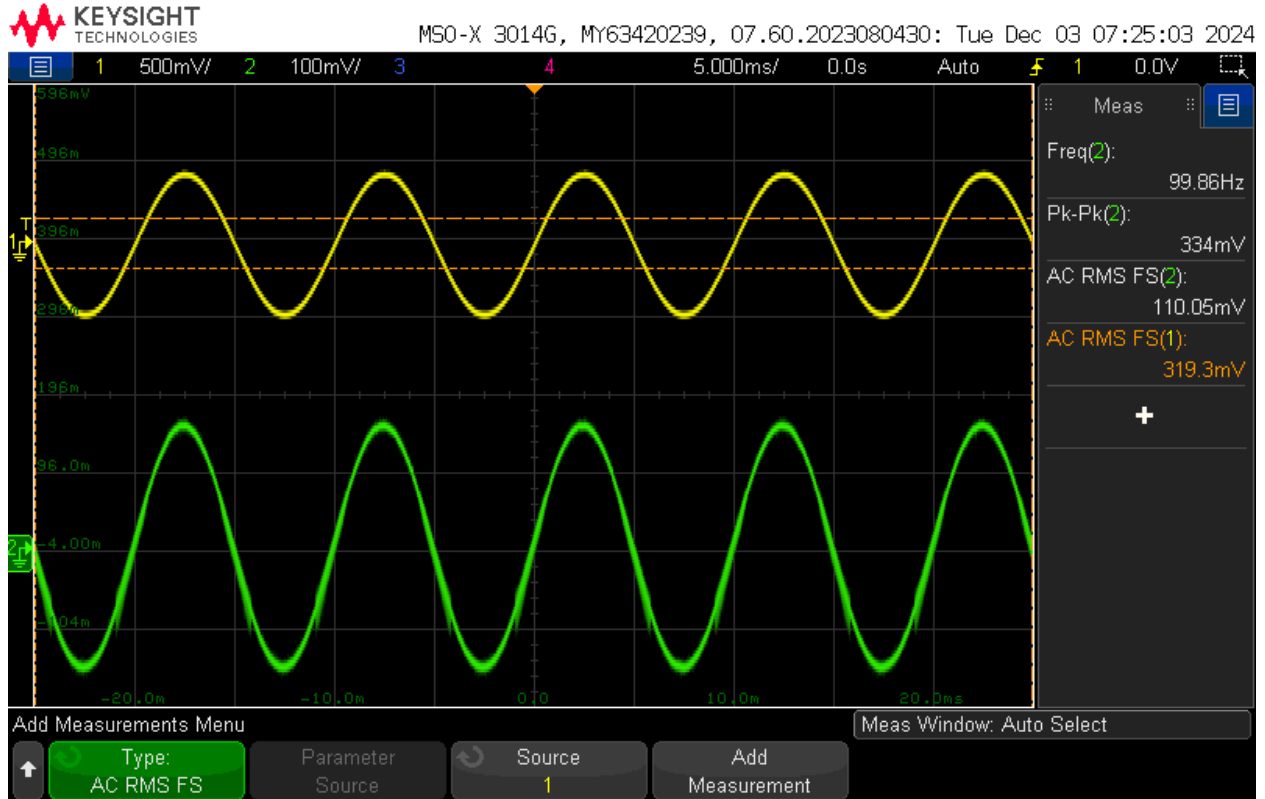


Figure 24: Ripple AC RMS Recording at 100 Hz Input

From this, the first voltage value to document is found, at 110.05 mVrms. Ultimately, this will be the $V_{RMS, Min}$ component of the ripple calculation.

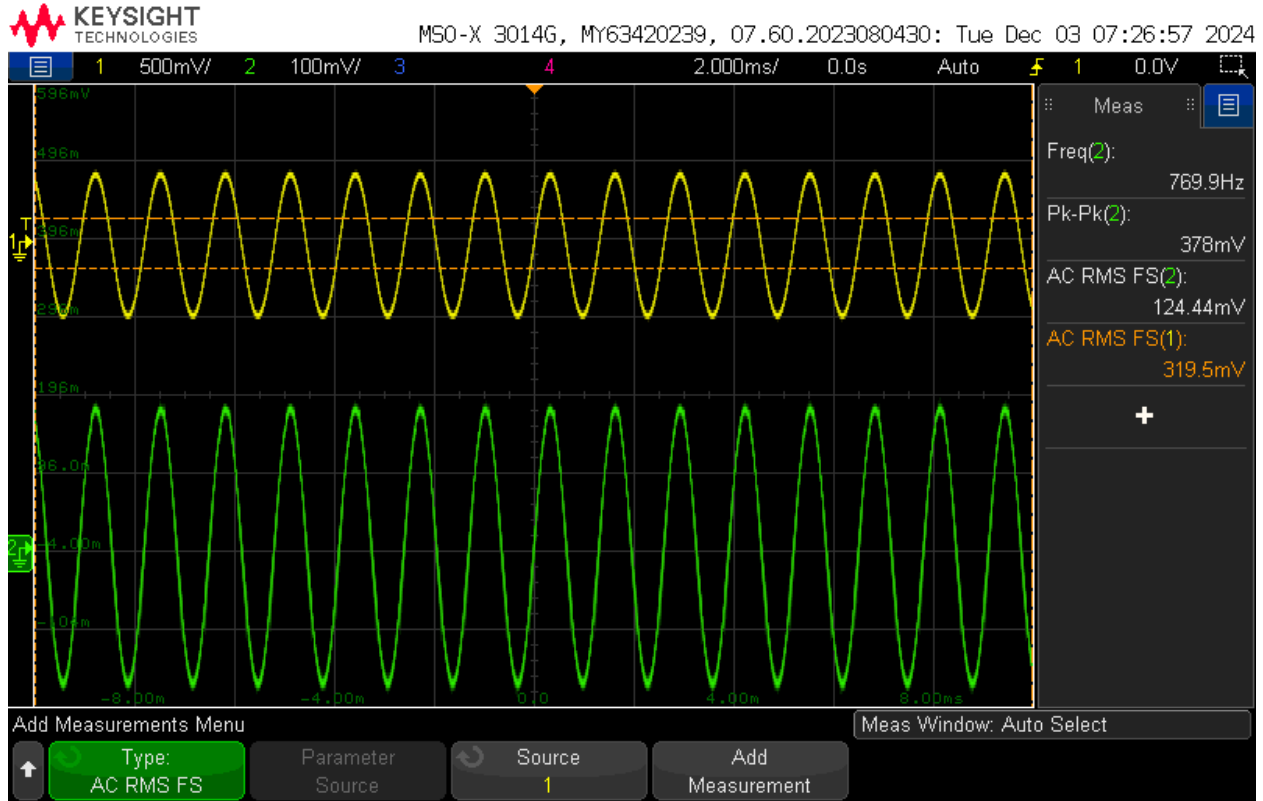


Figure 25: Ripple AC RMS Recording at 768.6 Hz Input

From this recording, a $V_{RMS,Max}$ value is found to be 124.44 mV. With these two values recorded, ripple is calculated:

$$Ripple = V_{RMS,Max} - V_{RMS,Min} = 124.44 - 110.05 = 14.39 \text{ mV}_{RMS}$$

This is less than the threshold value of 15 mV_{RMS} , so the ripple is within specification, and measurements on the summing amplifier are concluded.

4.2.2 Power Amplifier

The last and possibly most important component to test and confirm proper functionality of is the power amplifier. To test this, a power calculation is necessary using equation 4 below.

$$P = \frac{V^2}{R}$$

Equation 4: Power Calculation

Although frequency shouldn't play a role, this was confirmed by testing power output at 100Hz, 1kHz, and 10kHz just as before. With that said, only the final 10kHz measurement is recorded.

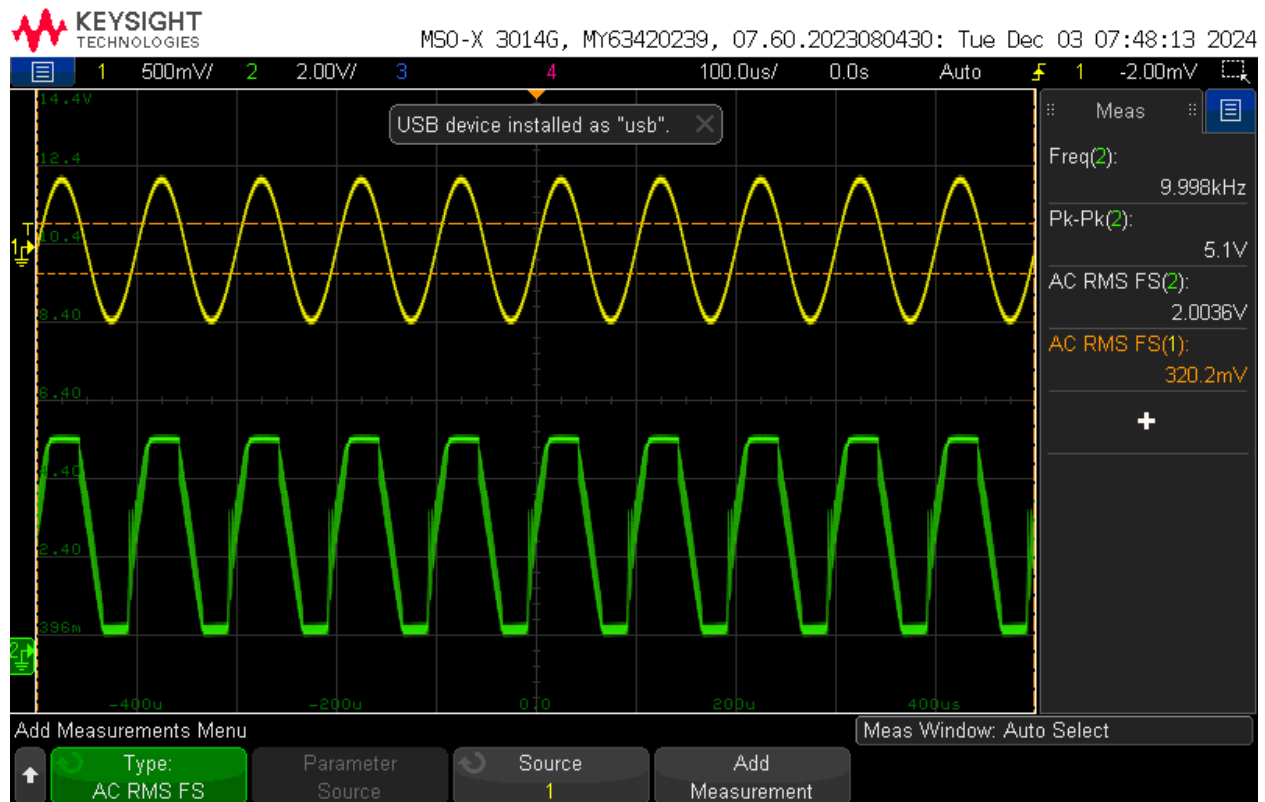


Figure 26: AC RMS Output At Maximum Master Gain

From Figure 26, the output voltage is noted to be 2.0036V. When this is plugged into the power calculation, 500W is the final output. It is worth noting that this is tested at a 6V input to the power amplifier, which is capable of handling 9V. With a higher input voltage, more power can be output. Having said that, as seen in table 4, the required power output is 400 W, so the circuit is functioning correctly.

To corroborate proper functionality across the entire bandwidth of the circuit, oscilloscope readings were taken at low, medium and high volume (toggled using the master gain knob). These can be found for analysis in figures 27-29.

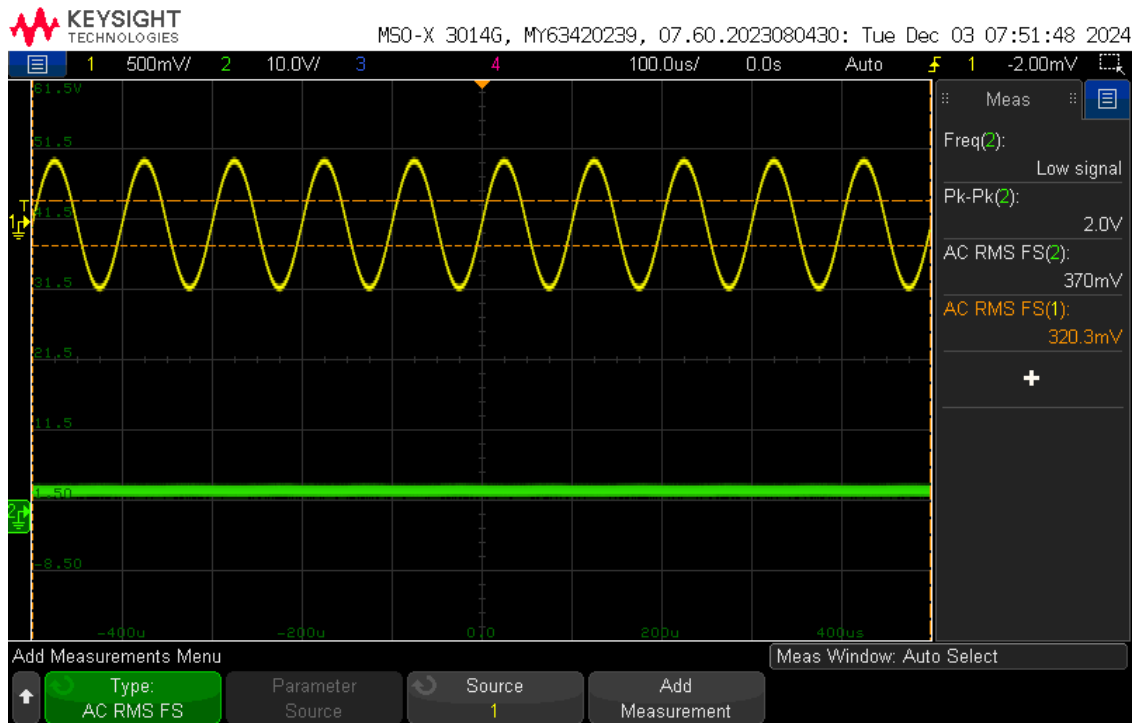


Figure 27: Low Master Volume Output with 320 mVrms input

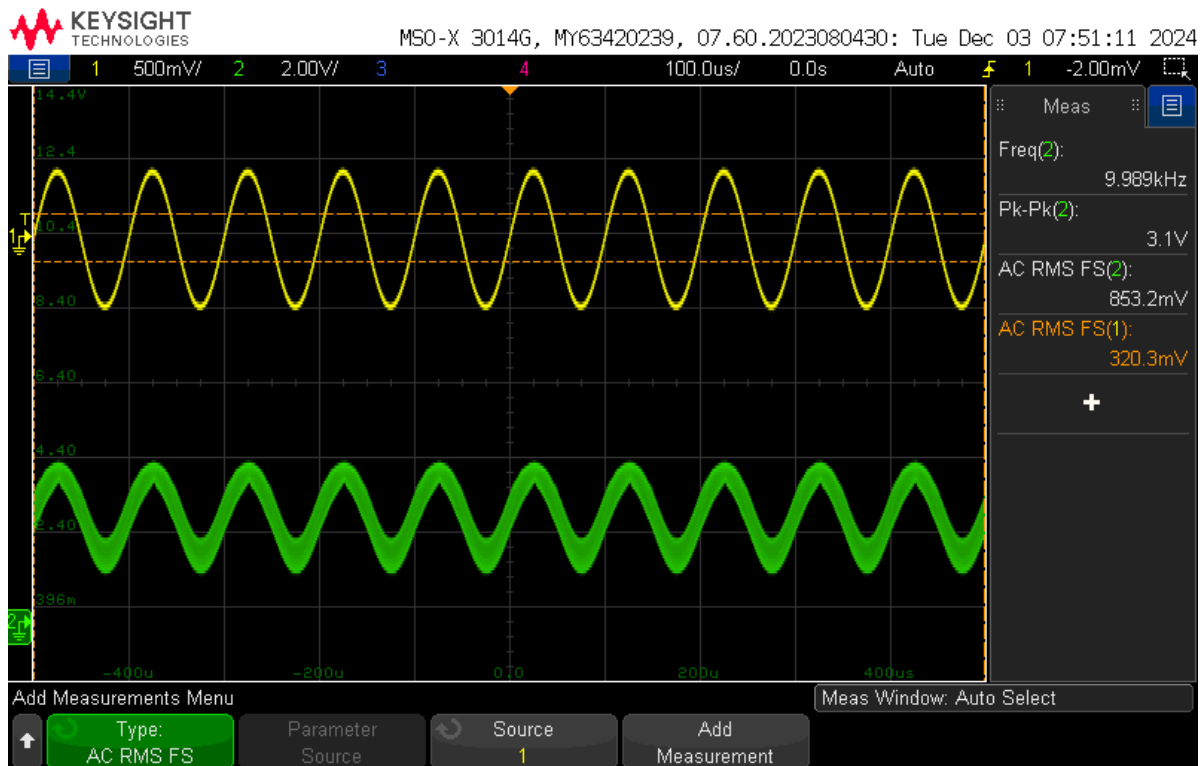


Figure 28: Middle Master Volume Output with 320 mVrms input

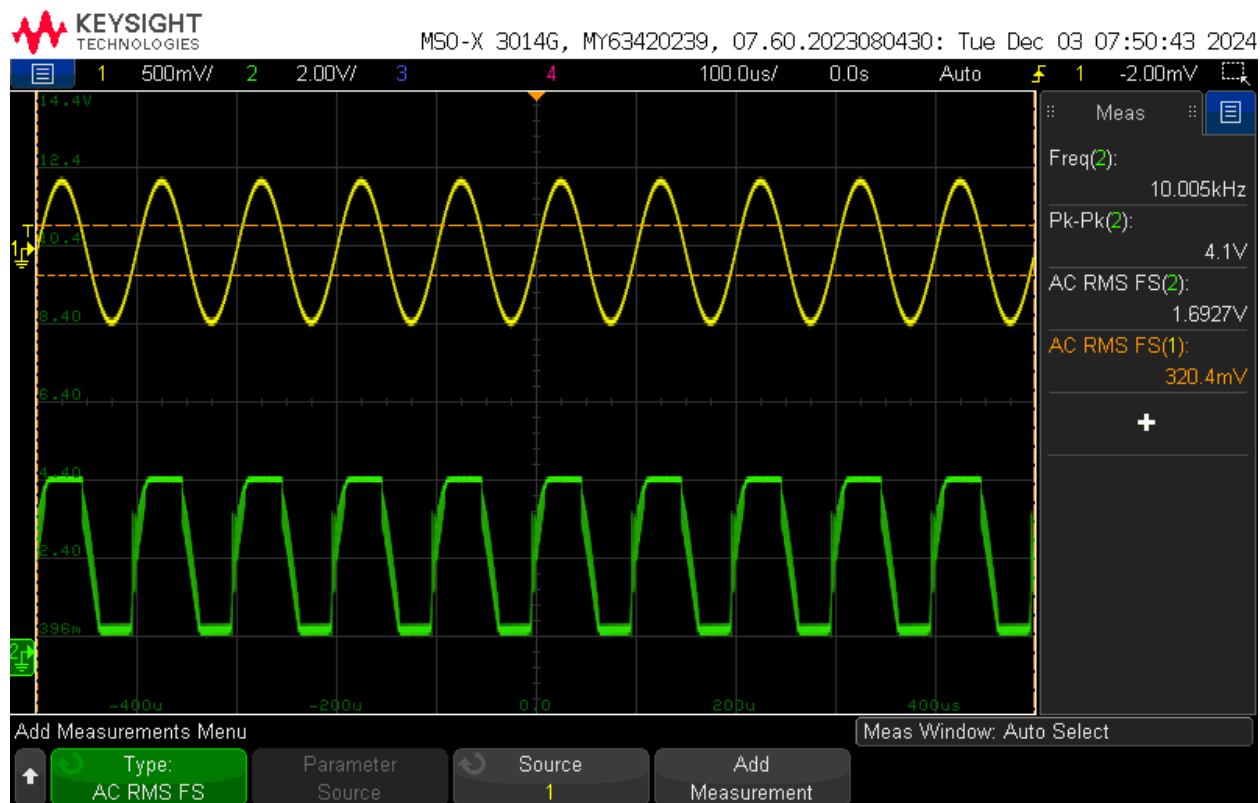


Figure 29: High Master Volume Output with 320 mVrms input

5. Conclusion

All things considered, this power amplifier circuit with integrated 3 channel equalizer has been thoroughly tested at each of its components, and functions properly. It also fulfils 100% of the requirements posed in table 4. The amplifier very effectively takes in a signal from a 3.5 mm auxiliary jack from a computer or other music source, and first splits it into 3 channels. These channels are organized by frequency band and each offer full individual variability. This allows direct and separate control of bass, mids, and trebles. From there, the signals are rejoined, and once more fed through a feedback loop providing “master” volume control. Finally, the signal is amplified with a gain of 20, and sent to the passive speaker where it is converted to sound.